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Accelerating Molecular Generation through Representation Alignment

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Declaration

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Abstract

Write a summary of the whole thing. Make sure it fits on one page.

Contents

Chapter 1

Introduction

The evolution of generative machine learning in scientific domains is defined by the shifting tensions between data scale, data quality, and structural priors. Our report examines a recent addition to this story, *representation alignment*, in two regimes for de-novo molecular generation — small molecules and protein backbones — and characterises its behaviour. In this introduction, we trace the narrative arc of research trends in molecular generation, and highlight the motivation for our work: the emerging need for data-efficient modelling in a world of prior-light models.

1.1 A history of trade-offs

The dominant ideas in generative modelling are *denoising diffusion* and its offshoot *flow matching* [1, 2], which found breakthrough success in the realm of image generation. This paradigm has since been adapted for de-novo design challenges across scientific fields, such as generating protein backbones [18, 19, 14], inorganic materials [21], and drug-like small molecules [15]. Training these models is typically expensive in both data and compute, with state-of-the-art image generators routinely training on billions of examples [35]. However, the corpus of available high-quality molecular data sits orders of magnitude below this. Experimental determination is slow and expensive, which means datasets are correspondingly small. For example, protein structure generators have largely been trained on at most half a million structures [14], which may explain why they have not yet matched the maturity of their vision-domain counterparts.

Historically, researchers compensated for the small scale of high-quality molecular data by encoding structural priors directly into model architectures [41]. SE(3)-equivariant architectures, for example, enforce the rotational and translational symmetries expected of molecules by construction [43]. But hard structural priors come with a computational cost, and the operations required to encode geometry are a bottleneck to scaling models.

A recent wave of research has moved entirely the other way: drop hard priors, use barebones non-equivariant transformer stacks, and rely on data scale and augmentation to teach the model what equivariance buys. This shift fits a broader trend in machine

learning on leveraging representation instead of model architecture to inject soft inductive biases for structured data. Vision Transformers [44] showed that the spatial-locality and translation-equivariance priors of convolutional neural nets (CNNs) can be learned rather than imposed, given sufficient scale. Graph Transformers like TokenGT [45] have made the same case for graphs. AlphaFold 3 [47] stripped many of the SE(3)-equivariant biases of AlphaFold 2 [46], replacing the structure module with a diffusion model operating on raw coordinates. Brehmer et al. [48] make the empirical scaling case directly: more data through a simpler model beats less data through a richer one. In our work, we examine Tabasco [15] and Proteina [14], which are flow-based generators built on largely vanilla transformers for small molecules and protein backbones respectively.

Given the difficulty of procuring data in scientific domains, data scale must necessarily come from non-experimental sources, which may be of lower quality. For example, recent efforts in protein generation have leaned on synthetic structures from folding-model predictions [14, 37] to expand training datasets into the tens of millions. However, even these enhanced corpora are arguably insufficient. Metagenomic databases catalogue over two billion natural protein sequences [39], and the theoretical sequence space at a median protein length [40] is hundreds of orders of magnitude greater still. Even at the frontier of the field's data scale, we sample but a sliver of the distribution we wish to model.

In summary, the field has traded hard structural priors for architectural simplicity, and curated data for synthetic scale. However, this trade-off dilutes the structural signal available to a model. Furthermore, even our best attempts at creating large datasets may not be large enough. These twin challenges amplify the need for techniques that make training still more efficient without sacrificing quality.

1.2 Representation alignment

But why is training diffusion-transformer architectures expensive in the first place? One answer comes from Yu et al. [5], who argue that these models suffer from a *representation bottleneck*. During training, a model is implicitly asked to learn two distinct tasks simultaneously: (1) representation: extracting semantically meaningful encodings from complex data; and (2) generation: constructing data from these encodings. The standard denoising objective alone may not provide sufficient signal for representation learning, leading to slow convergence and the large training budgets these models require in practice.

This burden falls harder on the prior-light models we study. In the absence of equivariant architectures, the training objective must now shoulder the additional responsibility of learning structural symmetries. Moreover, the scarcity of high-quality data compounds the problem. The model must learn complex geometry from data that only sparsely samples the true distribution, with no architectural scaffolding to fall back on.

One response to this problem is to supply representational structure from elsewhere. Instead of being entirely left to the training objective, it can be injected softly as an

auxiliary training signal. This is the idea behind Representation Alignment (REPA), which Yu et al. propose as a remedy. The technique adds a regularisation term that aligns a generator's intermediate hidden states with the embeddings of a frozen pretrained encoder. On image diffusion, they report roughly a 17.5× training speed-up against a strong baseline to achieve similar generation quality. REPA has since been extended beyond images to video [9], audio [10], and recently to molecular generation [7, 8], where it has been demonstrated but not systematically characterised.

1.3 Contributions

In this light, the core question we investigate in this report is: *can prior-light molecular generation models benefit from representation alignment with pre-trained models, in training efficiency and generation quality?* We aim to characterise when and why this works, including failure modes and limitations.

We document two lines of enquiry.

(i) *Does REPA work in the molecular domain?*: We conduct empirical studies on training flow-based generators with REPA in two regimes: Tabasco [15] for small molecules, and Proteina [14] for protein backbones. We find that REPA does not measurably improve Tabasco, but it improves training efficiency and advances the generation quality envelope for Proteina.

(ii) *When does REPA work?*: Following Singh et al. [6] in the realm of image generation, we develop a profiling framework to characterise the suitability of an encoder as a representation target. For every encoder, we measure its *information* content, the *relevance* of that content for the generation task, and the mathematical *conditioning* of its embeddings for the alignment objective.

We do not propose a new architecture or alignment loss, and our two studies are not a controlled experiment over the conditions that govern REPA's effectiveness. Instead, they are best read as paired case studies in regimes where those conditions differ.

The remainder of this report is organised as follows. Chapter 2 sets out the background and context needed to read the rest: flow matching, REPA, and molecular generation. Chapter 3 discusses what it means for a generative model to be "good" in our domain, and the tensions that question hides. Chapter 4 develops the encoder-profiling framework and uses it to predict where REPA may and may not help. Chapters 5 and 6 present the two paired studies, small molecules first and protein backbones second. Finally, we conclude in Chapter 7 with a discussion of whether what we learn generalises beyond molecular generation.

Chapter 2

Background and Related Work

In this chapter, we introduce the machinery the rest of this thesis depends on: flow matching (§2.1), representation alignment (§2.2), the de-novo molecular design pipeline (§2.3), and the two domains and models we work with (§2.4). The treatment is cursory by design. We intend only to guide the reader through this report, and we provide citations for omitted details.

2.1 Flow matching

The term *generative modelling* suggests *creating* new artefacts from scratch. However, it is more precisely the problem of *sampling* new artefacts from a complex data distribution accessible only through a finite training set. A common strategy here is to fix a simple distribution we can sample from cheaply — typically standard Gaussian noise $p_0 = \mathcal{N}(0, I)$ — and learn a mapping from it to the desired data distribution p_1.

Flow matching (FM) [2] realises this mapping as a continuous-time transport. The model designer defines a *probability path* between a noise sample and a data sample — a continuous interpolation between them — and the model learns to carry the noise sample along this path. Concretely, it parameterises a time-dependent *velocity field* $v_\theta(x, t)$ that gives the instantaneous direction of motion at each location x and time $t \in [0, 1]$. At inference, integrating the learned ordinary differential equation (ODE) $\dot{x}_t = v_\theta(x_t, t)$ from $t = 0$ to $t = 1$ pushes a fresh noise sample forward into a fresh data sample.

A common choice of probability path is the linear interpolant, $x_t = (1 - t)\, x_0 + t\, x_1$ for $x_0 \sim p_0$ and $x_1 \sim p_1$. The model's target is the *marginal* velocity field at (x, t): the average direction of motion over all noise-data pairs whose paths pass through that point. Supervising this marginal directly is intractable, since the contributing pairs are not known in advance. However, the key observation [2] is that the *conditional* velocity along the path of any specific noise-data pair is tractable: for the linear interpolant, simply $\dot{x}_t = x_1 - x_0$. Regressing the model on these conditional velocities, averaged over many sampled pairs, recovers the marginal field in expectation, and gives the training loss as a

mean-squared regression:

$$\mathcal{L}_{\mathrm{FM}} \;=\; \mathbb{E}_{t,\,x_0,\,x_1}\,\left\|v_\theta(x_t,t)-(x_1-x_0)\right\|^2 .$$

In practice, every training step samples a time $t \in [0,1]$ and a noise-data pair (x_0, x_1), computes the interpolant x_t, and takes a gradient step on this squared error.

Flow matching with this linear interpolant generalises score-based diffusion: the variance-preserving Gaussian-path objective of denoising diffusion models [1] is a special case of the FM family [2, 4].

Generating samples from a trained flow-matching model amounts to integrating the learned velocity field, often with a standard ODE solver. The field also admits stochastic (SDE) samplers that inject noise during integration. The noise amount is a tunable knob that broadens diversity at some cost to per-sample quality, with the deterministic ODE its zero-noise limit. The models we work with, Tabasco and Proteina, support both.

Conditional generation augments v_θ with context tokens — such as a class label, a sequence length, or a structural motif — with conditioning supplied at both training and sampling time. In this report, we focus on unconditional generation only.

For fuller treatments, see Lipman et al. [2, 4].

2.2 Representation alignment

Chapter 1 introduced the argument made by Yu et al. [5] about the *representation bottleneck* in diffusion-transformer models. It applies to flow-matching models too, by the equivalence above (§2.1). The flow-matching objective implicitly asks a model to learn two things at once: a useful representation of the data, and a velocity field that interprets this representation. This twin burden, they argue, leads to slow convergence and large training budgets.

Their proposed solution, Representation Alignment (REPA), addresses the first half of this burden by adding a regularisation term to the training loss. Figure 2.1 shows the construction. The regularisation pulls z_ℓ, the generator's intermediate hidden state at a chosen *alignment layer* ℓ, towards the embeddings produced by a frozen pretrained encoder f_{enc} run on the clean data sample x_1. The hidden state z_ℓ is a sequence of N token vectors. In the context of images, each token represents an image patch; in the molecular setting, each token represents an atom or residue that is part of the larger molecule (§2.3). A small trainable *projector head* h_ϕ — a three-layer multi-layer perceptron (MLP) in the original paper — maps z_ℓ into the encoder's feature space, so that $h_\phi(z_\ell)$ and $f_{\mathrm{enc}}(x_1)$ can be compared. This alignment is asymmetric. The encoder provides a fixed target throughout training, while the generator parameters θ and the projector ϕ are updated.

The alignment loss is the negated similarity, averaged token-wise, between projected and

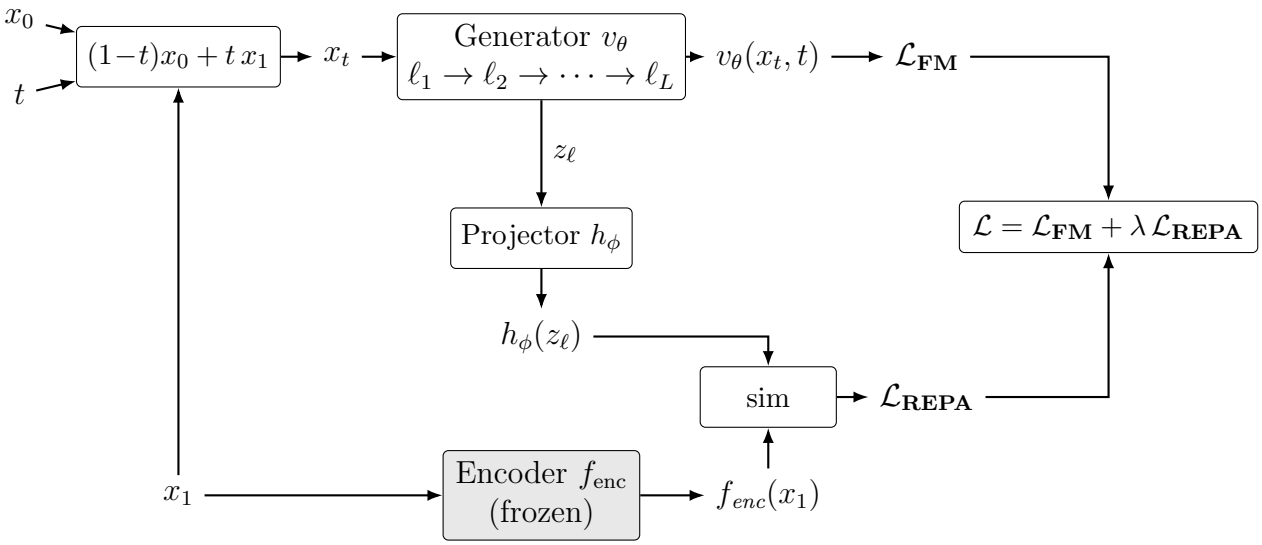

Figure 2.1: The REPA construction. A noised sample $x_t = (1-t)x_0 + t\,x_1$ is constructed at each training step from a noise sample x_0, a clean data sample x_1, and a sampled timestep $t \in [0, 1]$. The generator v_θ processes x_t through its sequence of layers $\ell_1, \ldots, \ell_L$ to produce the velocity field that feeds the flow-matching loss $\mathcal{L}_{\text{FM}}$. Separately, the clean sample x_1 is also passed through a frozen pretrained encoder f_{enc} (shaded). The generator's hidden state z_ℓ at the chosen alignment layer is mapped into the encoder's feature space by a trainable projector h_ϕ; the REPA loss $\mathcal{L}_{\text{REPA}}$ is the negated similarity between $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$, averaged across tokens. The total training loss $\mathcal{L}$ combines them with weight λ.

target embeddings:

$$\mathcal{L}_{\text{REPA}} \;=\; -\,\mathbb{E}_{t,\,x_0,\,x_1}\left[\frac{1}{N}\sum_{n=1}^{N}\text{sim}\big(h_\phi(z_\ell)_n,\; f_{\text{enc}}(x_1)_n\big)\right],$$

where n indexes the N tokens within a sample. In images N is constant, so a token-wise average matches a per-sample-then-batch average; for molecules N varies per sample, and the two diverge.

The final training objective is the flow-matching loss from §2.1 plus this term, scaled by a weight λ:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{FM}} \;+\; \lambda\,\mathcal{L}_{\text{REPA}}.$$

This construction exposes several design choices, some of whose effects we revisit in the ablations of Chapter 6:

- The encoder f_{enc}, since different encoders capture distinct aspects of the data.
- The alignment layer ℓ at which the projector attaches. Yu et al. found mid-network layers best for images, but this is only an empirical observation.
- The alignment-loss weight λ, which balances alignment against the generative objective. Yu et al. used 0.5 for images.
- The similarity metric sim. We adopt cosine similarity, which Yu et al. found more effective than mean-squared error.

- The averaging mode. Averaging within each sample weights every molecule equally, while pooling across the batch gives larger molecules proportionally more REPA signal.

We adopt the core REPA construction as in Yu et al. for this report. However, a small family of methodological variants on REPA has since emerged. Dispersive regularisers [12] replace the external encoder with an internal contrastive term, while flexible-target formulations [7] broaden which encoder features the loss attaches to. Separately, REPA-style alignment has been applied beyond the image setting, to video [9], audio [10], materials [11], and molecules [8]. We defer exploration of these variants to future work.

2.2.1 What makes for a good REPA encoder?

The encoder f_{enc} is among REPA's largest design choices, but what makes a good alignment target? Yu et al. [5] observed that ImageNet linear-probe accuracy correlates positively with generation performance, suggesting global semantic representation is the driving factor. More recent work by Singh et al. [6] reports the opposite after profiling a wider range of encoders: linear-probe accuracy does not predict REPA gain, while a measure of the spatial structure of patch-token representations does. Generation performance may therefore be driven by *local* feature geometry rather than *global* semantic strength. Encoder characterisation for REPA remains open, particularly in the molecular setting, and we pick up the thread in Chapter 4.

2.3 A primer on molecular generation

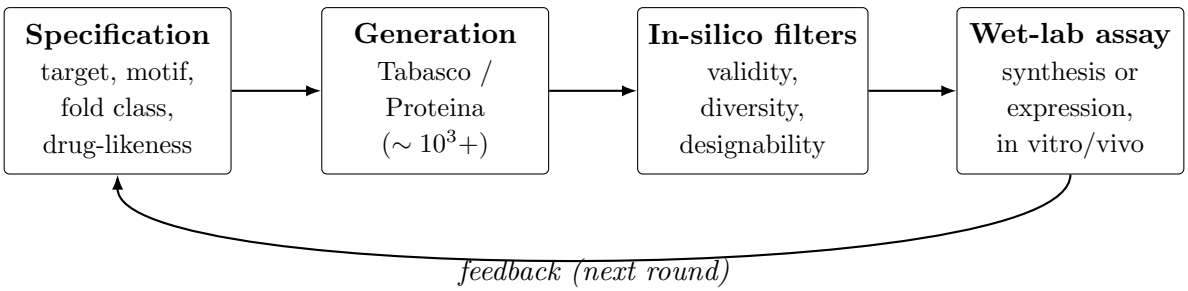

Figure 2.2: The de-novo molecular design cycle. An upstream specification drives a generative model to produce a candidate pool, which is funnelled through in-silico filters and then synthesised (or expressed) and assayed in the wet lab. Experimental results feed back into the next round of generation. Typical generation-step pool sizes are in the thousands to tens of thousands of candidates per round [18].

We now turn to the molecular setting, and the role generative models play within the broader de-novo molecular design pipeline. In a real-world scenario, generative models are rarely intended to produce ready-to-use artefacts. Instead, they act as high-throughput in-silico proposal engines in an iterative process of refinement and selection. This process, often called a *design campaign* [56], is summarised in Figure 2.2.

An upstream specification, such as a drug-likeness profile or a fold class, conditions a generative model to produce a pool of candidate samples, which are then funnelled through in-silico filters. For small molecules, these typically include checks for structural validity and physical plausibility [57], drug-likeness [60], and target docking; for protein backbones, designability and structural diversity are the dominant considerations [14]. The select few survivors are produced in a laboratory, through chemical synthesis (small molecules) or expression in cell lines (proteins), and evaluated using in-vitro or in-vivo assays. Experimental results then feed back into the next round of generation.

This pipeline framing is relevant to our discussion of evaluation metrics in Chapter 3. Several metrics we adopt, such as structural validity and designability, are not measures of intrinsic molecular quality, but proxies for specific in-silico filter steps.

2.4 Tabasco and Proteina

The two models we study — Tabasco [15] for small molecules and Proteina [14] for protein backbones — are both non-equivariant transformer flow-matching generators, designed to scale with data and model size rather than impose structural priors by construction.

Tabasco is motivated by the success of similar architectures in protein-structure generation, like Proteina, setting it apart from equivariant message-passing small-molecule generators such as EQGAT-diff [51], MiDi [52], and SemlaFlow [53]. In turn, Proteina is motivated by the pair-bias attention mechanism introduced in AlphaFold 3 [47], which distinguishes it from SE(3)-equivariant backbone generators such as RFdiffusion [18], FrameFlow [49], and Genie [50]. Despite their architectural minimalism, both achieve performance comparable or superior to equivariant baselines on quality evaluations.

2.4.1 Small molecules and Tabasco

In medicinal chemistry, a *small molecule* is a drug-like compound at the sub-900-Dalton scale [54]. In-silico, its structure is typically represented using two coupled features: a discrete *molecular graph* of typed atoms connected by typed bonds, and a continuous *conformer* placing those atoms in 3D Cartesian space. A flow-matching model must therefore handle both parameterisations jointly.

Tabasco [15] represents each molecule as a sequence of per-atom tokens, each carrying an atom-type embedding and a 3D coordinate, but with no explicit bond decoder. This treats the molecule as a point cloud of typed atoms rather than a graph, sidestepping edge modelling at the cost of having to recover bond structure post hoc from interatomic distances. Tabasco is trained on GEOM-drugs [38], a public dataset of drug-like conformers.

This point-cloud-of-atoms parameterisation is one of several, each with a different trade-off. *SMILES* and *SELFIES* string encodings discard the 3D conformer entirely. *3D voxel grids* preserve geometry but lose the graph and inflate the token count. *Equivariant point*

clouds, used by EQGAT-diff [51], MiDi [52], and SemlaFlow [53], keep both but require SE(3)-equivariant layers. See Duval et al. [42] and Wang et al. [23] for broader surveys.

2.4.2 Protein backbones and Proteina

A *protein* is a chain of amino-acid residues, ranging from around 50 to several thousands in length [55]; the largest known, titin, has approximately 34,000 residues. (Shorter chains are typically classified as *peptides* instead.) Each amino-acid residue contributes four *backbone* atoms (N, C_α, C, O) whose interactions determine the chain's *fold* geometry, and a *side chain* that determines its amino-acid identity.

A protein's *secondary structure* is its local pattern of α-helices and β-strands; its *tertiary structure* is the global three-dimensional fold those elements form when they pack together. A CATH code [64] (Class, Architecture, Topology, Homology) classifies this fold hierarchically. *Class* captures broad secondary-structure content (mostly-α, mostly-β, or mixed), *Architecture* the gross spatial arrangement of those elements, and *Topology* their specific connectivity.

For the structure-generation problem we study, it is customary to model only the backbone, with sequence identity recovered downstream by a separate *inverse-folding* model such as ProteinMPNN [25]. The *CA-trace* compresses this further, using only the C_α coordinate of each residue. It retains enough information to recover fold topology, at lower cost than the fuller *all-atom* representation. Proteina [14] adopts this C_α parameterisation, representing each backbone as a sequence of per-residue tokens carrying continuous C_α Cartesian coordinates. The models reported in the original presentation are largely trained on high-confidence synthetic structures from the AlphaFold Database (AFDB) [37].

Again, the CA-trace is one of several protein parameterisations. *Sequence-only* models such as the ESM lineage [24] operate on amino-acid strings, capturing rich functional priors but little 3D structure. *Full-atom* representations close the gap to physical realism at substantial compute cost. *Equivariant* parameterisations, such as torsion angles and oriented residue frames, encode SE(3) symmetries by construction and underlie the equivariant generators noted above, but have historically bottlenecked scaling. See Notin et al. [62] for a broader survey.

2.5 Open questions

This survey leaves open the two questions of §1.3. First, *does REPA work for molecules?* Existing molecular REPA work [7, 8] has not ablated over encoder choice, and protein backbone generation in particular remains unexplored. Second, *when does REPA work?* The encoder-characterisation work introduced in §2.2.1 has begun to ask which encoder properties predict REPA gain, but only for images, and we seek to explore this further. We take up both questions in Chapters 4–6.

Chapter 3

Evaluating molecular generation

What does it mean for a model to generate "good" artefacts? In this chapter, we outline a framework for evaluating molecular generation, and the tensions inherent to this endeavour. We argue that no single number suffices here, define the metric suites we use for small molecules (Table 3.1) and protein backbones (Table 3.2), and introduce the probe principle by which we measure a model's internal representations. We also introduce the theme of a *trade-off frontier* with these metrics. A gain on one axis may imply a loss on another (Chapter 6), so finding an acceptable balance, or better yet, pushing this envelope, is very useful.

3.1 What makes for a "good" molecule?

"Goodness" is usually task-specific, and a de-novo design pipeline (§2.3) asks several things of a generated candidate pool at once. A useful set must at least be *valid*: realistic enough to survive downstream filters; *diverse*: spread across the design space, not collapsed onto a few modes; and *designable*: amenable to synthesis. These axes trade off against each other. A model can buy validity by memorising a handful of safe scaffolds, at the cost of diversity; or buy diversity by spreading its mass thinly, at the cost of designability. No axis is dispensable, and maximising any one in isolation is uninformative, as these metrics are themselves proxies for the pipeline's real-world filter steps.

3.2 Representation and generation are multi-factorial and multi-scale problems

Moreover, the need for a suite of evaluation metrics reflects how molecules are represented in-silico. Molecules present a stark contrast with images, which originated the generative machine learning methods we adopt (§2.1).

In a 2D image, *geometry* and *content* are coupled in one representation. The pixel grid is at once an object's shape and its identity. There is a single contiguous feature space, so

one self-supervised encoder can serve as a near-complete description of it. This coupling is what lets image generation lean on a single strong encoder, and lets image quality be summarised, to a first approximation, by a single Fréchet distance in that encoder's feature space [59].

However, molecules pull geometry and content apart, and into more than two pieces. A molecule is at once a discrete graph of typed atoms drawn from a periodic-table-sized alphabet, and a continuous 3D geometry. This is a discrete–continuous duality with no clean analogue in the pixel grid. The two concepts are only loosely tied: one bonding graph admits many conformers, and similar geometries can host different atom types. Geometry is itself organised hierarchically from local environments up to global shape. Indeed, the taxonomy used to describe protein structure makes this explicit, running from primary sequence through secondary-structure elements to the tertiary fold.

As a consequence, both representation and generation quality are *multi-factorial* and *multi-scale* in this domain. No single encoder captures every factor of content and geometry at every scale at once, as we see in Chapter 4. Moreover, no single metric can score a generated set along all those factors at once. A model can match the distribution of folds while getting local geometry subtly wrong; it can place atoms in valid positions while collapsing the diversity of shapes it produces. These are independent failures, so mode collapse is harder to reason about than in the image domain, where one number suffices to track what goes wrong.

The metric suite we propose in this chapter is an attempt at handling this complexity. We decompose quality along the factors the domain pulls apart, and interpret metrics holistically to understand the complete picture. For each domain we report two complementary families: *representation quality*, which scores what the model encodes internally, and *generation quality*, which scores its outputs. We present them in that order throughout the report, since representation is what REPA's mechanism acts on (§2.2) and generation is the outcome we hope it improves.

3.3 The linear probe principle

One way to understand what a model represents internally is the *linear probe*. We freeze the model, extract its hidden states for clean inputs, and fit a linear regression to a known structural label. If the label is recoverable by this simple probe, that information is considered to be present and accessible in the representation; if not, it is considered absent or deeply entangled. This follows standard practice in image generation, where encoders are often characterised by ImageNet linear-probe accuracy [5]. We use this mechanism in two places: on the frozen encoders we might align to (Chapter 4), and on the generator trunk itself (Chapters 5 and 6).

The form of the read-out follows the label. For real-valued targets, we fit a regression and score recoverability by held-out R^2. For categorical targets, we fit a logistic classifier and

score top-1 accuracy.

We outline the small-molecule targets we regress against in §3.4, and the protein suite in §3.5. Each probe maps to a necessary condition for a useful alignment target. The tasks span content and geometry from local to global scales, so we interpret a representation that scores well across the whole battery as genuinely carrying holistic information.

3.4 Small-molecule metrics

Table 3.1 collects every small-molecule metric we use, following the conventions of recent 3D molecular generators [15, 51, 53].

Representation quality. This asks what structural information the generator has learned to encode internally. We apply the probe principle of §3.3, regressing mean-pooled per-atom features onto four global RDKit descriptors [61] — molecular weight, LogP, ring count, and rotatable-bond count — and score recoverability by held-out R^2.

Generation quality. This asks if the generator produces good artefacts. For small molecules, quality metrics fall into three groups: per-molecule *structural quality* (validity, connectivity, uniqueness), set-level *coverage* (diversity, novelty), and *distributional realism* (the Fréchet ChemNet Distance, FCD [58]). The first group guards against malformed or repeated molecules; the second against mode collapse and memorisation; FCD, the small-molecule analogue of the image-domain Fréchet Inception Distance [59], asks whether the generated set as a whole resembles real molecules. As we explore in Chapter 5, the structural-quality metrics are particularly amenable to saturation by a strong generator. This does not mean the problem is solved; rather, it is a limit of the yardstick itself.

3.5 Protein-backbone metrics

Protein backbones are larger and more variable than small molecules (Chapter 2), and evaluating them is correspondingly more involved. Table 3.2 collects all the protein metrics we use. We present representation first, then generation, and close by offering our own framework for how the generation metrics should be grouped and read together.

Representation quality. We apply the probe principle of §3.3 for three tasks adapted from the literature, spanning local to global structure: per-residue inverse-folding recovery [28] for the local chemical environment; backbone dihedral error [27] for low-level geometry; and CATH fold classification [27] for global fold. The first two are scored per residue (top-1 accuracy and mean absolute error), while the fold probe is a logistic classifier on mean-pooled per-chain features. Because the three are largely orthogonal, improving all of them at once is strong evidence of holistic representation quality.

Representation alignment. This asks not what the generator encodes, but how geometrically close it sits to a frozen encoder. We measure it with CKNNA [65], a mutual-nearest-neighbour kernel alignment, following the original REPA analysis [5] and a recent

Metric	Definition	Content	Score
Representation quality — probe recoverability of RDKit descriptors [61]			
MolWt ↑	Molecular weight	Global size	R^2
LogP ↑	Octanol–water partition coefficient	Lipophilicity	R^2
Rings ↑	Ring count	Topology	R^2
RotBonds ↑	Rotatable-bond count	Flexibility	R^2
Generation quality			
Structural quality			
Validity ↑	Fraction sanitising to a valid molecule once bonds are inferred	Malformed geometry	Fraction
Connectivity ↑	Fraction forming a single connected component	Fragmentation	Fraction
Uniqueness ↑	Fraction not duplicating another generation	Trivial repetition	Fraction
Set coverage			
Diversity ↑	Mean pairwise dissimilarity across the generated set	Mode collapse	Fraction
Novelty ↑	Fraction of the set absent from the training data	Memorisation	Fraction
Distributional realism			
FCD ↓ [58]	Fréchet ChemNet Distance: Fréchet distance over the whole set in a property-prediction network's feature space	Distributional realism	Distance

Table 3.1: Small-molecule metrics (Chapter 5), following the conventions of recent 3D molecular generators [15, 51, 53]. Arrows give the good direction.

molecular extension [8]. We use it as a diagnostic lens on how REPA acts rather than a quantity we optimise.

Generation quality. Measures of generation quality draw on two lineages. The *canonical* ones — designability, and various counts of novelty and diversity — score or tally individual generations, and predate modern generative models. The *model-based* ones, introduced to this domain by Proteina [14], borrow the idea from image generation of comparing a generated set against a reference in a learned feature space. We headline on one from each: designability as the canonical measure of per-sample quality, and the Fréchet Protein Structure Distance (FPSD) as the model-based measure of set-level distributional realism. We outline these two metrics, add a secondary-structure metric of our own, and then set out a framework that organises them all.

Designability. This metric estimates the fraction of generated backbones that are realisable as real proteins. It involves an in-silico round-trip through the protein design loop (§2.4.2). Each backbone is inverse-folded with ProteinMPNN [25], a model trained to predict an amino-acid sequence that would fold into a given backbone, to propose candidate sequences. These sequences are then re-folded with ESMFold, a model trained to predict 3D structure from sequence. A backbone is termed designable when the self-consistency

Metric	Definition	Content	Score
Representation quality			
IF top-1 ↑	Per-residue inverse-folding sequence recovery [28]	Local chemical env.	Top-1 acc.
Dihedral MAE ↓	Per-residue backbone dihedral-angle error [27]	Low-level geometry	MAE
CATH-A top-1 ↑	Per-chain fold-architecture classification [27] (Table A.2)	Global fold	Top-1 acc.
Representation alignment			
CKNNA ↑ [65]	Per-residue mutual-nearest-neighbour kernel alignment, trunk vs. encoder	Trunk–encoder similarity	Score
Generation quality			
Quality			
FPSD ↓	Fréchet Protein Structure Distance: distance over the whole set in a structural encoder's (GearNet) feature space	Distributional realism	Distance
Des ↑	Fraction of generated backbones that are designable	Physical realisability	Fraction
Tertiary structure — whole set (T-W)			
fJSD-A ↓	Jensen–Shannon divergence vs. reference CATH-A distribution	Fold-distribution match	Divergence
fS-A ↑	Spread of generated folds across CATH-architecture classes	Whole-set fold diversity	Count
Tertiary structure — designable subset (T-D)			
#Clust ↑	Number of TM ≥ 0.5 structural clusters	Designable diversity	Count
pwTM ↓	Mean pairwise TM-score	Designable diversity	TM-score
Secondary structure — whole set (S-W)			
ssJSD-2D ↓	Jensen–Shannon divergence of the (helix-frac, strand-frac) distribution	Secondary-structure match	Divergence
Secondary structure — designable subset (S-D)			
ssJSD-2D ↓	As above, on the designable subset	SS match (designable)	Divergence
E%des ↑	Mean β-strand (E) fraction	β-content	Fraction

Table 3.2: Protein-backbone metrics (Chapter 6). Arrows give the good direction; full per-metric results appear in Tables 6.7 and 6.9.

root-mean-squared deviation (scRMSD) between the generated backbone and its re-folded prediction falls below 2 Å.

One caveat affects interpretation: designability is an in-silico proxy, not ground-truth physical realism, so it inherits the biases of the folding models it relies on. The inverse-folding and folding steps share model lineage with AlphaFold2 [46], and hence with the synthetic AlphaFold Database (AFDB), whose structures are themselves AF2 predictions. Generators trained on AFDB therefore score systematically higher on designability by construction, as we explore in Chapter 6.

FPSD. The Fréchet Protein Structure Distance [14] is the protein analogue of the imaging FID. First, it embeds the generated and reference sets using an external encoder: in this case, a GearNet [26] model trained on a fold classification task. Then, it models each set of embeddings as a Gaussian, and reports the Fréchet distance between the two — a scalar that drops to zero as the generated distribution matches the reference in both mean and covariance. Because the encoder reads fold-level geometry, FPSD scores how well a generated set reproduces the reference distribution of folds. As a distributional distance, FPSD also rewards matching the spread of the reference, not just its centre. It therefore carries some diversity signal of its own, which is why it and our dedicated diversity metrics often move together.

ssJSD-2D. We introduce this secondary-structure metric in our report. Like FPSD and fJSD, it is a *distributional* metric, scoring the generated secondary-structure distribution against a real reference. Proteina already measures this, but only at the level of mean helix and strand content, which is blind to a model that matches the average composition while collapsing its spread. In contrast, ssJSD-2D is a Jensen–Shannon divergence over the full joint (helix-fraction, strand-fraction) distribution against the reference, so it rewards reproducing the whole spread of secondary-structure content, not just its centre (Appendix A.8).

A two-axis taxonomy. We propose reading all these metrics through a framework that organises them by structural and distributional relevance. Two axes do the work: *tertiary* versus *secondary* structure, and the *whole set* versus the *designable subset*. The result is a grid of metric groups, which the result tables in Chapter 6 lay out as supercolumns. *Tertiary-whole (T-W)* asks whether the distribution of global folds matches a reference over all backbones; its flagship is FPSD, alongside fJSD (a Jensen–Shannon divergence between the generated and reference fold-class distributions) and a fold-spread score (fS) [14]. Both fJSD and fS are defined at each level of the CATH hierarchy (§2.4.2); we headline on the architecture level (fJSD-A, fS-A). *Tertiary-designable (T-D)* asks the same diversity question, but only of backbones that pass designability, via cluster count and pairwise TM-score (pwTM). The metric differs from T-W's by design: the designable subset is far smaller than the whole set, so a distributional measure like fS is less reliable over it, whereas pwTM is a pairwise average that stays robust at small sizes. *Secondary-whole (S-W)* and *secondary-designable (S-D)* apply ssJSD-2D over the same two populations.

A caveat on noise. We also note that generative metrics are noisy, so a gap between two point estimates is not necessarily a finding. We report bootstrap intervals and seed sweeps where possible to guard against this.

3.6 The designability–diversity trade-off

A central dynamic in protein generation pits the designability number against the diversity of the designable subset (T-D). A model can trivially raise its designability number by collapsing onto a few reliable, highly foldable modes, but this shrinks the diversity of that designable subset. Pushing designability up therefore tends to push designable-subset diversity down, and vice versa. This dynamic is noted regularly in the literature [14], and is typically modulated by a choice of sampler noise.

Ultimately, how does this trade-off help us compare models? In our studies, we distinguish between two phenomena. The first is whether a model merely occupies a different spot on the same envelope, in which case the optimal choice is use-case specific. The second is whether a model pushes the envelope itself — playing the same trade-off game, but at a higher level, winning on one axis without conceding the other. This makes a model canonically better.

These complex dynamics further motivate the question: *what makes an encoder a good teacher of representational information in this domain?* We examine this question in Chapter 4, which informs the experiments we present in Chapters 5 and 6.

Chapter 4

Encoder targets and profiling

What properties make an encoder a good target for representation alignment? This question is unsettled in the image domain, with global semantic quality and local spatial structure both proposed as the driver (§2.2.1).

In this chapter, we ask this question in the molecular domain. Earlier, we discussed how molecular representation is multi-factorial and multi-scale (§3.2), which means that no single pre-trained encoder is likely to capture all of it. An encoder is typically optimised for one task in isolation, and sees a molecule through that lens. For example, an interatomic-potential model reads local geometry; a fold classifier reads global shape; a sequence model reads chemical identity. The encoder we align to therefore selects which factor REPA is most likely to transfer.

Our profiling work in this chapter makes this concrete. We build a suite of diagnostics to assess what information an encoder contains, how much of it is relevant to the generative task, and how well conditioned its embeddings are. We use this to profile seven candidate encoders across small molecules and proteins, and then use the results to guide which encoders to align with in the studies of Chapters 5 and 6.

4.1 A profiling protocol

We profile every candidate encoder with the protocol summarised in Table 4.1. Each encoder is run frozen on a sample of structures from our training sets: 200–2,000 molecules (QM9 and GEOM) for the small molecule encoders, and 200 proteins (PDB) for the protein encoders. This yields one embedding vector per atom or residue — tens of thousands of vectors in each case — and every diagnostic is computed from these per-token embeddings.

We ask three things of each encoder, and a good target needs all three.

Information asks: *what does the encoder contain?* This estimates the upper bound on what REPA can add to a model. Alignment cannot teach anything the encoder does not itself hold. For a 3D generator, the axis that matters most is 3D geometry. We

Diagnostic	What it captures	How we compute it
Information	*what the encoder encodes — an upper bound on REPA's transfer*	
3D geometry	sensitivity to 3D pose	cosine self-similarity across conformers (molecules) or after a 1 Å coordinate perturbation (proteins)
Token identity	atom or residue type (high = a lookup the molecular generator already has; off-task for the protein generator, which is not given residue identity)	linear-probe accuracy
Relevance	*what the generator can match without improving its representation*	
Saturation	the embeddings barely vary, so a constant prediction already matches them	a high *floor*: the average cosine similarity of each token embedding to the dataset mean
Cheatability	the variation that exists is content the generator is already fed (for molecules; for proteins this identity variation is read as off-task, not cheatable)	a high *lift* from identity alone: the rise a REPA-form 3-layer MLP gains from a one-hot of the atom or residue type (cosine-similarity loss, 80/20 split)
Off-task risk	the remaining variation is some factor the generator does not need, rather than the 3D geometry it does	a low floor and low lift, read against the 3D-awareness probe
Conditioning	*whether the embedding is well-formed*	
Effective rank	dimensions actually used	RankMe [32] (Roy–Vetterli effective rank)
Sparsity, dead dims	inactive dimensions	fraction of exact zeros; permanently inactive dimensions
Norm scale	numerical conditioning	mean embedding ℓ_2 norm

Table 4.1: The profiling diagnostics, grouped by the three properties: information, relevance, and conditioning.

measure geometry-awareness by how far the embedding moves under a change in 3D pose: across distinct conformers of the same molecule for the molecular encoders, and after a 1 Å coordinate perturbation for the protein encoders, where only a single structure is available. We also report token identity, since a high score there suggests that an encoder is performing an identity lookup that the generator may already be capable of.

Relevance asks: *how much of a REPA target can the generator match without improving the representation it is trying to learn?* A generator does not need to learn all the information contained in an encoder. Indeed, during training, it is fed most of the relevant *content* of the molecule directly: the normalised token index always, and, for molecules, a

one-hot of the atom type (the protein generator is given position but not residue identity; see §4.3). What it must acquire is the 3D *geometry*, which it sees only as noisy coordinates it is being trained to denoise. Hence, a REPA target is relevant only insofar as matching it forces the generator to improve that geometric representation.

A target can fail this in two ways:

- *cheatable*: the target is trivially reconstructible from inputs the generator already holds. So the model can saturate alignment loss without actually learning a better representation.

- *off-task*: matching the target does require representation learning, but what it requires is not what we need. For example, since the protein generator is not given residue identity, matching an identity-encoding target would force it to infer sequence from backbone. This is a genuine learning problem, but a different one from generating geometry, so the pressure may not assist the task we care about.

We estimate the cheatable fraction directly. We fit a three-layer MLP, the same projector that REPA uses (§2.2), to reconstruct embeddings from cheap features that withhold all 3D geometry: first *identity* alone (a one-hot of the atom or residue type), then *identity+position* (Table 4.1). Separately, with no projector fit, we also score the *constant* dataset mean, which is the optimal input-free prediction. We read this constant baseline as the *floor*, and the rise of the fitted rungs above it as the *lift*.

Read together, the floor and lift sort an encoder into one of three configurations:

- high floor suggests *saturation*. The embeddings barely vary, so a constant prediction already matches them. There is little signal to align to.

- low floor, high lift suggests *cheatable*. The embeddings vary, but the cheap inputs reconstruct that variation, so it is largely an identity lookup using data the generator is already fed.

- low floor, low lift is *promising*. However, a low lift only says the variation does not stem from the cheap identity we tested. It does not positively affirm that it comes from the geometry a generator needs. We interpret this against the information probes to tell a geometric encoder from one varying for some *off-task* reason.

Conditioning asks: *is the embedding mathematically well-formed enough to be a target?* For example, an embedding whose variance has collapsed onto a few directions (a low effective rank), or that is mostly zero, is a bottleneck. Too little structure passes through for the alignment to transmit anything, no matter the information or the relevance.

We apply this protocol to molecule encoders first (§4.2), then protein encoders (§4.3), and draw together what it implies in §4.4.

Encoder	Dim	token-ID	3D	Floor	Lift	Sparsity	Verdict
CheMeleon (2D graph)	2048	1.00	1.00	0.43	+0.04	93.8%	too sparse
MACE (3D potential)	192	1.00	0.998	0.86	+0.005	0%	saturated

Table 4.2: Molecule encoder profiles. Neither encoder shows promise. *token-ID*: atom-type linear-probe accuracy (trivially 1.00 for both; uninformative). *3D*: cosine self-similarity across conformers (lower = more 3D-sensitive). *Floor*: constant-prediction cosine (high = saturated). *Lift*: the rise a same-form MLP gains above the floor from cheap inputs. *Sparsity*: fraction of exact zeros.

Encoder	Dim	AA-probe	3D	Floor	Lift	RankMe	Verdict
GearNet	512	0.137	0.27	0.43	+0.009	256	usable
random-init	512	0.127	1.00	0.95	+0.003	100	saturated
ProteinMPNN	128	0.327	0.73	0.84	+0.014	85	weak but usable
ESM2-650M (L33)	1280	0.998	N/A	0.67	+0.053	1031	off-task†
PW-GearNet	3072	0.928	0.86	0.71	+0.013	208	poorly conditioned
MC-GearNet-Edge	3072	0.146	0.77	0.86	+0.002	12	collapsed

Table 4.3: Protein encoder profiles. GearNet and ProteinMPNN show promise. *AA-probe*: residue-identity linear-probe accuracy (high = an identity lookup the generator already has). *3D*: cosine self-similarity after a 1 Å perturbation (lower = more 3D-sensitive; N/A for sequence-only ESM2. *Floor*: constant-prediction cosine (high = saturated). *Lift*: the rise a same-form MLP gains above the floor from cheap inputs. *RankMe*: Roy–Vetterli effective rank out of *Dim* [32]. †ESM2's lift is the largest here but is sequence context, not 3D: off-task.

4.2 Small molecule encoders

We profile two publicly available pre-trained encoders, summarised in Table 4.2 (full values in Appendix A.10). They bracket a useful contrast. One sees the 2D bonding graph, the other sees 3D geometry. Neither, the framework finds, is a clean alignment target.

CheMeleon [31] is a directed message-passing network (a ChemProp D-MPNN) that passes messages along the bonds of the 2D molecular graph; it never sees coordinates. Its embeddings are well spread (a low floor), but it fails on two other counts. First, it is 93.8% sparse, so alignment runs over a shifting, ill-conditioned subspace. Second, it is conformer-invariant by construction, so its lift is 2D bond-graph context, which may not aid a 3D generator.

MACE [30] is an equivariant message-passing network built on many-body atomic interactions, pretrained as an interatomic potential to predict energies from 3D structure. The model's embeddings are mathematically well-formed. However, its energy pretraining captures local environments over global pose, so its 3D signal is weak, and a high floor of 0.86 suggests it is mean-saturated.

4.3 Protein backbone encoders

Proteins offer a wider choice of pre-trained encoders. We profile five candidates plus a random-initialised reference, summarised in Table 4.3 (full values in Appendix A.10).

GearNet [26], our lead candidate, is a message-passing network. It passes messages over a residue graph wired by spatial and sequential proximity. The checkpoint we profile is pre-trained for CATH fold classification and released with Proteina [14]. It is genuinely 3D-aware (a 1 Å perturbation drops the cosine self-similarity to 0.27) and barely a residue-identity lookup (AA-probe 0.137). The embeddings are spread, and the remaining headroom is likely geometric. The random-init row suggests that the architecture alone gives a near-constant, low-rank embedding, and the training significantly improves on this.

ESM2 [24] offers an instructive contrast, as a transformer-based protein language model trained by masked-residue prediction over sequence alone. It produces the largest lift in the field, yet that lift is sequence context. It ignores coordinates by construction, and its last layer flattens almost entirely to residue identity (AA-probe 0.998 against GearNet's 0.137). This information is real but likely unhelpful to a backbone generator.

ProteinMPNN [25] is a message-passing network over a nearest-neighbour backbone graph, pretrained for inverse folding (predicting sequence from structure). It has the cleanest conditioning of any 3D-aware encoder we profile, but its geometric signal is significantly weaker than GearNet's (a 1 Å perturbation leaves the cosine self-similarity at 0.73, against 0.27), and it has a relatively high floor. We characterise this encoder as weak but usable: its 3D signal is faint, but clean and genuinely geometric.

The remaining two are GearNet-Edge variants. PW-GearNet [26, 27], pretrained on backbone torsions, is only weakly 3D-sensitive and poorly conditioned: its effective rank uses just 208 of 3072 dimensions. MC-GearNet-Edge [26] is collapsed entirely, with 95% of its variance in a single dimension.

Of the five candidates, only GearNet clears all three diagnostics, with ProteinMPNN also showing promise. Each of the others fails a different gate.

4.4 Informing our experiments

Our profiles are measured before any alignment training, so the expectations they set are genuinely a-priori, not post-hoc rationalisations. We use them to guide which encoders our studies will showcase. For small molecules (Chapter 5), we expect no measurable REPA gain. For protein backbones (Chapter 6), we expect that GearNet and ProteinMPNN are most likely to help. The following two chapters put these ideas to the test.

Chapter 5

REPA on small molecules

Having profiled candidate encoders for their suitability as a representation alignment (REPA) target in Chapter 4, we now put that analysis to the test, beginning with small molecules. We ask: *can REPA accelerate or improve the training of Tabasco [15], a flow-matching transformer-based generator of 3D drug-like small molecules?*

Our headline finding is that no REPA variant measurably improves, or accelerates the training of, Tabasco's small-molecule generation. On structural-quality metrics, the baseline is already at ceiling. On FCD, the one metric with headroom, the REPA variants perform no better than baseline. Nor do they reach any level of quality faster: the baseline leads from the beginning of training. We explore *why* in §5.3, where two properties of this regime — a saturated evaluation surface and the absence of a usable teacher — emerge as an explanation and seed the contrast with proteins (Chapter 6).

5.1 Experimental setup

5.1.1 Dataset

We train on GEOM-drugs [38], a dataset of drug-like molecular conformers at the sub-900-Dalton scale. It holds roughly 1.4M molecules, split ∼80/10/10 into 1,142,099 training, 146,411 validation, and 146,798 test molecules. We model heavy atoms only, and augment each sample with eight random rotations. Each molecule is represented as a small point cloud of typed atoms, typically fewer than fifty, as outlined earlier (§2.4.1).

5.1.2 Models

The baseline is the unmodified Tabasco flow-matching transformer (§2.4.1): a 3.7M-parameter non-equivariant model that denoises 3D atom coordinates and atom types jointly. Our REPA variants add a trainable projector that aligns the trunk's hidden states to a frozen encoder, evaluated on the clean (t=1) molecule (§2.2). The trunk carries two hidden-state streams, one for coordinates and one for atom types. We concatenate them and pass the pair through a single shared projector, so the alignment acts on one

Property	Value
Base architecture	
Backbone (*trunk*)	Tabasco: 3.7M-param, 16-layer non-equivariant transformer (hidden dim 128, 8 heads, SiLU, cross-attention)
Inputs (what the trunk sees)	per-atom token index + atom-type one-hot + noisy 3D coordinates
REPA alignment	
Encoder targets	CheMeleon (2D, 2048-d) · MACE (3D, 192-d)
Injection depth ℓ	final (last) layer
Combination mode	additive $\mathcal{L}_{\text{gen}}+\lambda\mathcal{L}_{\text{REPA}}$ (default) · tradeoff (convex) $(1-\lambda)\mathcal{L}_{\text{gen}}+\lambda\mathcal{L}_{\text{REPA}}$
Projector	single shared 3-layer MLP
Similarity metric sim	cosine similarity
Averaging mode	per atom
Alignment weight λ	0.8
Training & sampling	
Learning rate	2×10^{-3}
Batch size	256
Weight decay	none
EMA decay	0.999
Sampler	SDE (Langevin) interpolant, 100 integration steps
Prior noise scale	1.0 (unit Gaussian)
Langevin schedule	$1/(t+0.01)$, disabled for $t \geq 0.9$
White-noise scale	0.01

Table 5.1: Model and training configuration. This chapter presents two encoder targets; the tradeoff combination mode is examined in §5.3.

combined representation. It enters the loss as a plain additive regulariser, leaving the flow-matching objective untouched.

Encoder ablations. The two encoders we study come from the profiling of Chapter 4, and span a useful contrast. *CheMeleon* [31] is a 2D molecular-graph encoder. It is conformer-invariant, so it sees only the bonding graph and not the 3D pose. Meanwhile, *MACE* [30] is a 3D interatomic-potential encoder. It is geometry-aware, but, as the profiling showed, its output is a narrow descriptor of local atomic environments. Hence, the former carries rich 2D-graph information, while the latter encodes 3D geometry.

5.1.3 Metrics

Representation quality. We use the small-molecule probe battery of Chapter 3 (Table 3.1): a linear ridge regression from mean-pooled per-atom features to four RDKit descriptors [61], scored by held-out R^2. We apply it to both the frozen encoders, to see what each REPA target carries on its own, and the generator trunk, to see what the model itself comes to encode, and rely on it in the diagnosis of §5.3.

The probe set comprises 1,000 molecules from the GEOM-drugs validation split. We fit the regression on 800 of these and report R^2 on the held-out 200.

Generation quality. We report the standard small-molecule generation metrics of Chapter 3 (Table 3.1): validity, connectivity, and uniqueness (structural quality); diversity and novelty (set coverage); and the Fréchet ChemNet Distance (FCD), the one distributional axis with headroom.

We compute every metric on a set of 1,000 molecules sampled from the trained model using the SDE interpolant described in Table 5.1. We mimic the molecular size distribution of GEOM-drugs in this set, and use the training set as the reference for our FCD metric.

5.2 Results: no measurable gain

No REPA variant meaningfully improves on the baseline. As outlined in Table 5.2, the baseline pushes validity, connectivity, and uniqueness to their ceiling, and the REPA variants follow suit. This is perhaps to be expected, since these are largely properties of bond topology, which a strong generator reproduces almost perfectly. Essentially, there is no headroom on these metrics for improvement.

On FCD, the one metric with room to move, both tested variants are worse. The baseline records the lowest (best) FCD at 5.61; CheMeleon lands at 7.43 and MACE at 6.81 (Table 5.2). We caveat this by noting that each variant is a single run, not a multi-seed average, and we do not have per-set FCD intervals to bound the gaps. Nevertheless, both point estimates favour the baseline.

There is no acceleration in generation quality. REPA's headline result in the image domain is a training speed-up: the same generation quality reached in far fewer steps (§2.2) [5]. No such speed-up appears here. On every axis, the REPA variants and the baseline are indistinguishable from the first few thousand steps (Figure 5.1).

The unequal training does not make this comparison unfair. The baseline trained roughly twice as long as the REPA variants ($\sim$152k steps over 33 epochs, versus $\sim$69–73k for the two REPA runs) for our final point estimates. We stopped the REPA runs early on purpose. With the quality metrics saturated and FCD flat or worse, there was no evidence that further training would change the verdict. Nor did shorter runs cost the variants anything on the saturated axes, which always plateau within the first few thousand steps.

5.3 Discussion

The null result suggests this regime offers neither headroom nor usable teachers. Under the original hypothesis, REPA [5] alleviates a representation bottleneck and therefore accelerates improvements on generation quality. However, this needs two conditions. First, there must be *headroom*: a gap in what the generator represents or produces, for the alignment to fill. Second, there must be a *usable teacher*: an encoder that encodes representational information that differs from what the trunk already has and transfers to the generative task. In this regime, neither condition holds.

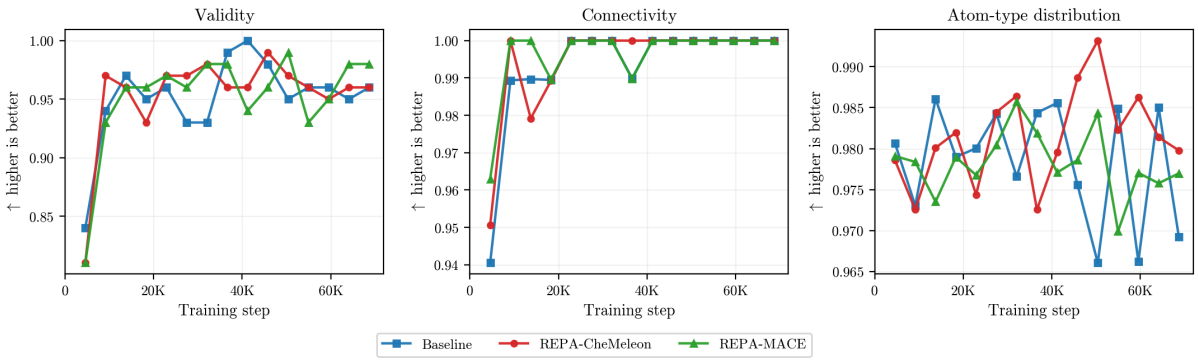

Figure 5.1: **REPA does not accelerate generation quality over the baseline.** Generation quality on GEOM-drugs (100 molecules sampled per epoch during training), plotted over training step across the range all three runs reach. The baseline and both REPA variants track together and reach near-ceiling on every axis within the first few thousand steps, leaving no headroom for alignment to fill. The metric is the in-training estimate, so values differ slightly from the final-checkpoint evaluation of Table 5.2.

Model	**Steps**	**Valid.**	**Conn.**	**Uniq.**	**Div.**	**Nov.**	**FCD ↓**
Baseline	151k	0.980	0.998	1.000	0.886	0.966	5.61
REPA-CheMeleon	73k	0.972	0.999	1.000	0.882	0.961	7.43
REPA-MACE	69k	1.000	0.995	0.996	0.884	0.977	6.81
REPA-MACE (tradeoff)	69k	1.000	0.999	1.000	0.883	0.982	6.32

Table 5.2: **REPA brings no measurable generation quality gain over the baseline.** The quality metrics are saturated, sitting at or near their ceiling for baseline and variants alike. On FCD, the only metric with headroom, every REPA run is worse. The MACE *tradeoff* row replaces the additive REPA loss with a convex one (§5.3). (1,000 sampled molecules; higher-is-better except FCD (↓). *Steps* is the training step at evaluation; the baseline trained ∼2× longer, but the quality metrics plateau early, so the unequal steps do not make the comparison unfair (§5.2). One run per variant.)

The first condition fails on the metric surface. The generation quality axes are saturated, as §5.2 showed. The representation side is saturated too — the trunk already encodes what our descriptors capture — so this regime happens to be closed on both.

The second condition fails on both encoders, for opposite reasons (Table 5.3). From our linear probes, we learn that CheMeleon decodes the descriptors almost perfectly in isolation (R^2 up to 0.99). But the baseline trunk *already decodes them well* (0.64–0.92), because they are 2D-graph quantities and the trunk sees the graph. CheMeleon's signal therefore overlaps with what the model already has. Moreover, FCD ultimately scores the inferred 2D graph — the very information the trunk already holds — so even on the one metric with headroom, CheMeleon adds nothing new.

MACE presents the contrary problem. It is genuinely 3D, but it encodes only a narrow, energy-tuned slice of *local* atomic chemistry. The global descriptors we probe sit outside that slice, so they are largely undecodable from its mean-pooled embedding (R^2 near zero, save a weak 0.31 on LogP). Its signal differs from the trunk's, but does not transfer.

Source	MolWt	LogP	#Rings	#RotBonds
Frozen encoders (the REPA targets in isolation)				
CheMeleon (2D)	0.993	0.988	0.995	0.989
MACE (3D)	0.022	0.307	0.081	−0.014
Dummy	0.822	0.160	0.432	0.345
Generator trunk (the model's internal representation)				
Baseline	0.920	0.713	0.742	0.636
REPA-CheMeleon	0.943	0.669	0.736	0.606
REPA-MACE	0.935	0.703	0.704	0.582
REPA-MACE (tradeoff)	0.955	0.730	0.732	0.666

Table 5.3: **Linear-probe R^2 reveals little representational difference between REPA and the baseline.** The baseline trunk already captures what CheMeleon learns, while MACE's embedding is too narrow to decode these descriptors. Hence, neither makes for a usable REPA target. (R^2 for four RDKit descriptors, on held-out molecules; §5.1.3.)

To calibrate the scale, note that an untrained random-weight encoder (Dummy) already decodes molecular weight at $R^2 = 0.82$, simply because weight tracks atom count. MACE falls below even this floor on three of the four descriptors.

Forcing the signal in does not rescue it either. As a final test, we replaced the additive regulariser with a convex *tradeoff* loss, $(1 - \lambda) \cdot \mathcal{L}_{\text{gen}} + \lambda \cdot \mathcal{L}_{\text{REPA}}$. This deliberately down-weights the flow-matching loss to force the model to adopt the encoder's representation. Even so, the effect on generation is negligible. MACE under the tradeoff loss still trails the baseline on FCD (6.32 versus 5.61; Table 5.2). Its only mark is on the representation, where it is the only variant to beat the baseline on a majority of the descriptor probes (Table 5.3), but only by a small margin. Even when forced into the architecture, MACE leaves only a tiny trace in representation, and nothing in generation.

When is REPA actually useful? Our results sharpen the question on REPA's effectiveness rather than settling it. Under our setup, REPA does not help because the regime offers neither headroom nor a usable teacher. This is not evidence that REPA cannot help with molecular generation in principle. Indeed, we do not claim to have exhausted the REPA design space, and there may be other configurations that work well. But it does suggest the question has no single answer. Molecular generation is a broad domain of distinct regimes, and whether REPA helps may be decided within each, not across the whole. Read that way, REPA is not a single intervention, but a family of them.

Practically, we must look for a regime within the domain that satisfies these two conditions. Protein backbone generation presents an intriguing possibility on both fronts, and that is where we turn next.

Chapter 6

REPA on protein backbones

In this chapter, we expand the scope of our study on representation alignment from small molecules to much larger ones: proteins. In Chapter 5, our efforts using REPA to generate small molecules yielded little demonstrable improvement, because the Tabasco [15] baseline already saturated most measures of generation quality, and because the encoders used were likely unhelpful. However, de-novo backbone design represents a more challenging task, as the data distribution of proteins is significantly larger, and the geometry of these structures is correspondingly more complex. In this context, we ask: *can REPA accelerate or improve the training of Proteina [14], a flow-matching transformer-based protein backbone generator?*

Our headline finding is that REPA accelerates convergence on the two key measures of protein generation quality. This holds across both experimental and synthetic training-data regimes (Figure 6.1). The one exception is designability on synthetic data, which the baseline already saturates under standard sampling (§6.1.1). This chapter characterises *what* happens and attempts to understand *why*. We note that our objective here is to test whether the mechanism transfers at all, not to search for the optimal REPA configuration.

Reading figures and tables in this chapter. Unless otherwise specified, figures plot data from the PDB-trained $n \leq 256$ data-regime (§6.1.1); we display the mean over three sampling seeds, with shaded bands showing the min/max. (Figure 6.1 spans both data regimes, and Figure 6.3's bands are bootstrap intervals.) In tables, the baseline is usually an absolute value and each REPA row a Δ from it; green marks a change in the metric's good direction and red a change against it; darker green signals the best entry in a column; small $|\Delta|$ are left uncoloured.

6.1 Experimental setup

We integrate REPA into Proteina [14], a 60M-parameter non-equivariant flow-matching generator of Cα protein backbones (§2.4.2). Following §2.2, we attach a trainable projector

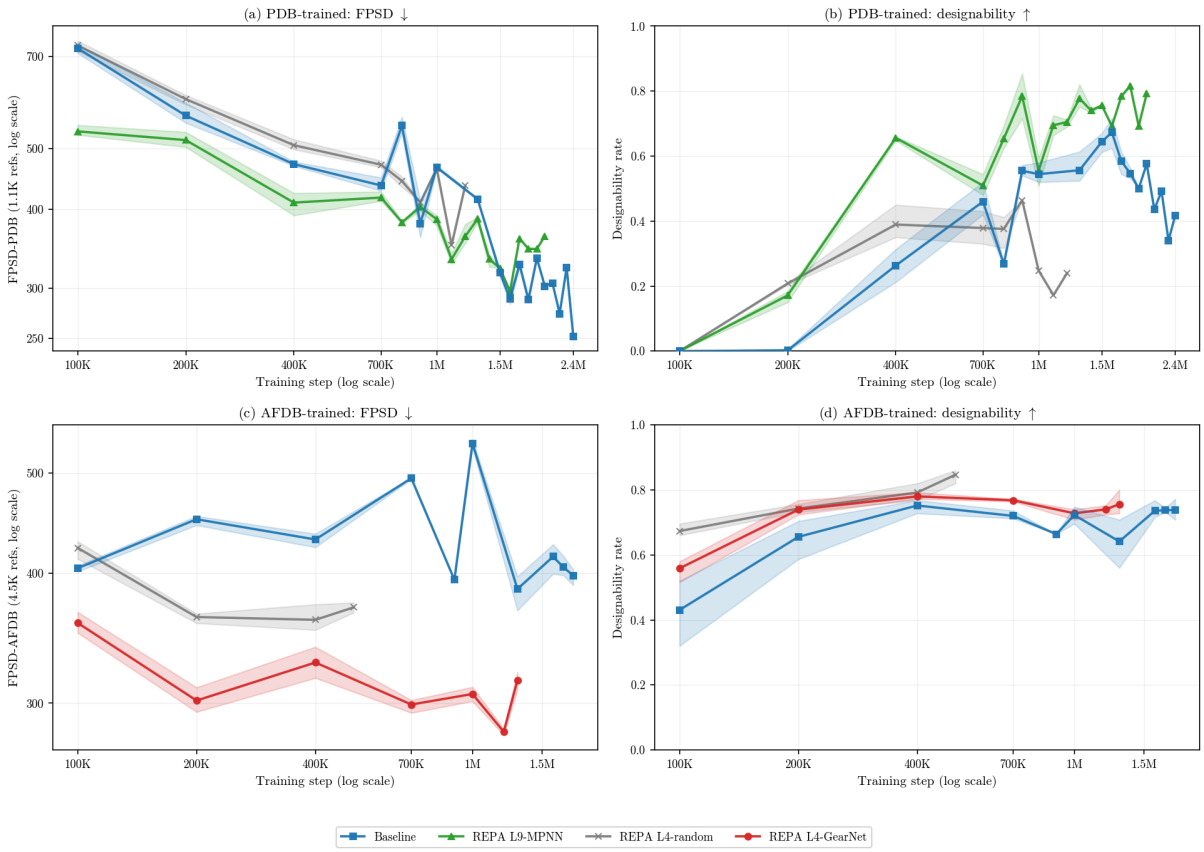

Figure 6.1: **REPA accelerates FPSD in both data regimes, and designability on experimental (PDB) data.** The random-encoder control is grey. PDB retains a persistent designability gap, but its baseline appears to catch up on FPSD after ≈1M steps; we examine this in §6.2.3.

at a chosen layer of the generator trunk (the transformer stack) that aligns the model's hidden states to a frozen external encoder.

6.1.1 Datasets

We train in two data regimes that mirror the data-scarcity context motivating this report (Chapter 1). The Protein Data Bank (PDB) [36] is the high-quality experimental source, but it is relatively scarce, containing on the order of hundreds of thousands of structures. The AlphaFold Database (AFDB) [37] is the synthetic alternative containing hundreds of millions of AF2-predicted structures, but it is almost entirely unverified. We treat the datasets as two ends of a quality–scale axis. (We use the manually-curated Swiss-Prot subset of AFDB as a reproducible stand-in for the Foldseek-clustered TrEMBL subset on which Proteina was trained in the original presentation [14]. This corpus contained data that has since been pruned from AFDB, and is hence inaccessible.)

Following Proteina [14], we train on the subset of chains with at most 256 residues; we represent this length-regime by $n \leq 256$. (We note that this represents only a property of the training datasets, not a bound on the length of the proteins that models trained on these datasets can generate.) Following this filter, the two training sets are comparable

in size (PDB: 273,771 / 3,190 / 323 train/val/test chains; AFDB: 229,670 / 4,521 / 235). Appendix A (Table A.1) contains further details on dataset composition. During training, we augment each backbone with a single random global rotation per sample.

Different datasets in this domain carry different biases. Trends in designability offer a clear example, as training on AFDB raises the in-silico-designability floor well above PDB at every setting we tested (§6.2.3). This is the designability-proxy lineage effect of Chapter 3: the test re-folds with models sharing AlphaFold2's lineage, so AlphaFold-like backbones are plausibly re-folded most confidently. We also note that both datasets feature some amount of structural similarity across training and evaluation subsets. We attempt to control for this in our evaluation protocol, as noted in §6.1.4.

6.1.2 Models

The base Proteina model comprises a stack of ten transformer layers (zero-indexed) that feeds a flow matching objective. Our experiments add a REPA loss to this design, as outlined in Table 6.1.

Encoder and injection-depth ablations. Our narrative concentrates on the two encoders identified as most viable in Chapter 4: *GearNet* [26], a structure encoder pre-trained for CATH fold classification (checkpoint released with Proteina [14]); and *ProteinMPNN* [25], which encodes each residue's local inverse-folding environment. Each encoder is attached to the trunk in two layer-depth configurations: layer 4 (middle layer) and layer 9 (last layer). The remaining encoder and depth combinations are collected in Appendix A.4 (Table A.3). Figures presented in this chapter will typically feature the best performing model to serve as an illustrative example.

Separating regularisation from mechanism. A random-weights GearNet attached at layer 4, identical in architecture but never trained, serves as a control architecture in this chapter. We use this model to separate whatever REPA achieves merely due to the regularisation effect of adding an auxiliary loss from what it achieves through the encoder's *learned* structural knowledge. Crucially, a trained encoder at the same layer 4 still outperforms the random one, so the learned-versus-random gap is not an artefact of injection depth (Appendix A.4).

Hyperparameters. Our defaults largely follow the original image-domain REPA recipe [5].

6.1.3 Metrics

The original hypothesis that motivated REPA for imaging [5] was that alleviating a *representation* bottleneck would improve *generation* quality. We seek to understand if the same mechanism applies in the molecular domain as well. Hence, along with estimating the quality of a model's generated samples, we are also interested in what the model comes to *represent* internally. To this end, we measure three families of metrics, as outlined below: representation quality, representation alignment, and generation quality (Table 3.2).

Property	Value
Base architecture	
Backbone (*trunk*)	Proteina (CAFlow): 60M-param, 10-layer non-equivariant Cα transformer (token dim 512, 8 heads)
Inputs (what the trunk sees)	per-residue token index + noisy Cα coordinates (no residue identity)
REPA alignment	
Encoder targets	GearNet (512-d) · ProteinMPNN (128-d) · random-GearNet (control)
Injection depth ℓ	layer 4 (middle) · layer 9 (last)
Projector	3-layer MLP, hidden dim 512
Similarity metric sim	cosine similarity
Averaging mode	per residue
Alignment weight λ	0.5
Training & sampling	
Learning rate	10^{-3}
Batch size	24 ($n \leq 256$)
Weight decay	none
EMA decay	0.999
Sampler	SDE, noise scale $\gamma = 0.45$
Integration steps	400 (step size $dt = 0.0025$)
Self-conditioning	on
Diffusion schedule	$g_t = 1/t$, log-spaced time steps

Table 6.1: Model and training configuration. This chapter presents two encoder targets and two injection depths. Other ablations are deferred to Appendix A.

Representation quality. We use the three-task protein battery of Chapter 3 (inverse folding, dihedral error, CATH fold classification; Table 3.2). Because the three probes target largely orthogonal capabilities — local chemical environment, local geometry, and global fold — improving all of them at once is strong evidence of holistic representation quality. Among the CATH levels we headline on architecture (CATH-A): Class is too easy (top-1 saturates) and Topology too sparse (too many buckets, too few chains). We report all three for completeness, the patterns holding across them, and motivate the choice in Appendix A (Table A.2).

Representation alignment. This measures how close the trunk sits to a structural encoder, via CKNNA (§3.3; protocol below). As in the original REPA analysis [5] and a recent molecular extension [8], we use it as a diagnostic — a mechanistic lens on *how* REPA might produce its representation gains — rather than a quantity we optimise or build claims on.

Generation quality. This is what ultimately matters, and we report it through the two-axis taxonomy of Chapter 3 — tertiary/secondary structure crossed with whole set/designable subset — laid out as supercolumns in our result tables: per-sample quality, then T-W, T-D, S-W, and S-D (Table 3.2). We report two headline metrics: designability and FPSD. We use this taxonomy to analyse REPA's effect on both the whole generated

distribution and the designable subset, along with the trade-off the latter implies.

Most of these metrics follow Proteina [14]; the one we add, ssJSD-2D, is defined in Chapter 3 (full description in Appendix A.8). As we explore in §6.2.5, REPA improves distribution match and whole-set diversity together, but may trade off tertiary-structure diversity *within* the designable subset.

6.1.4 Evaluation protocol

Representation probing. Following the probe principle of §3.3, we fit probes to a frozen checkpoint's hidden states on clean inputs (t=1): the inverse-folding and dihedral probes per residue, the CATH probe per chain on mean-pooled representations. IF is a 20-way classification, CATH-A a ∼40-way one.

Each probe is fit on 1,000 training chains. We evaluate the per-residue probes (inverse folding and dihedral) on a *blinded* set of ∼325 proteins (∼43k residues) held to under 30% sequence identity to all training data, which removes both model- and probe-side leakage. The per-chain CATH probe needs many distinct chains spread across fold classes, which that small slice cannot supply, so we relax it to a *probe-clean* evaluation on a larger validation set of ∼3,190 chains. We detail this scheme, and the model/probe leakage distinction it controls for, in Appendix A.3, and read rank-order and Δ-from-baseline throughout this chapter.

Representation alignment. Earlier, we outlined how CKNNA scores how closely two representations share their nearest-neighbour structure (§3.3). We compute it on clean (t=1) data from a fixed held-out batch, between the hidden states at every trunk layer and three domain-relevant encoders: GearNet, ProteinMPNN, and ESM2 [24]. We use k=10 neighbours over 10,000 residues, and bracket each value with a subsample-without-replacement bootstrap (50 draws at 80%; sampling with replacement would inflate the score through duplicate neighbour pairs). The absolute CKNNA values we record are small, an order of magnitude below image-domain REPA [5], so we read the direction and layer structure of the effect, and its separation from the baseline, not its raw magnitude.

Generation sampling. Our generation evaluation follows the Proteina protocol [14], at a reduced sample budget.

For whole-set evaluations, we sample 1,125 backbones per checkpoint — 125 at each length in $\{50, 75, \ldots, 250\}$ — and compute metrics over all of them. Depending on the training regime, we use matching PDB or AFDB reference statistics.

For designability evaluations, 50 backbones at each length $\{50, 100, 150, 200, 250\}$ are inverse-folded with ProteinMPNN (8 sequences each), re-folded with ESMFold, and called designable when the self-consistency RMSD is below 2 Å. The designable-subset metrics (T-D, S-D) are then computed on the backbones that pass.

With this experimental testbed in place, the rest of the chapter investigates what the REPA construction changes with respect to baseline, and how that helps improve gener-

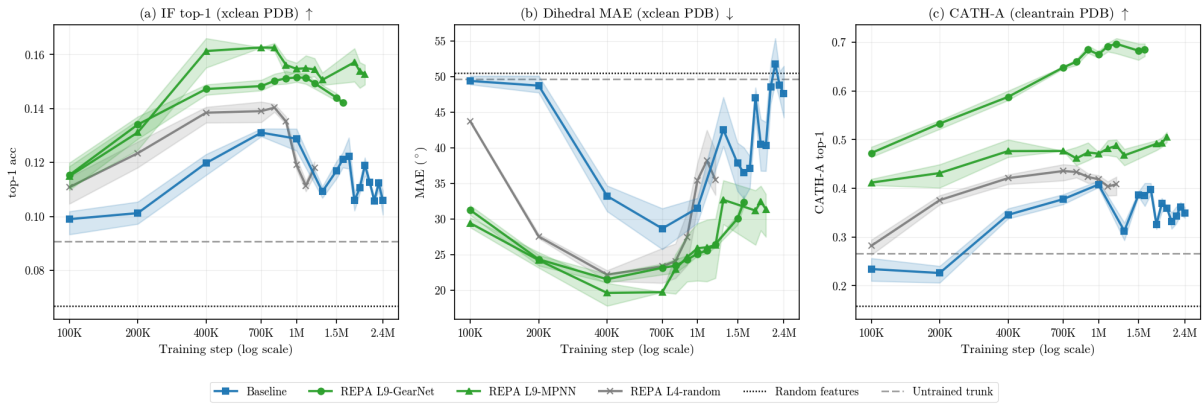

Figure 6.2: **REPA offers a mechanistic effect that lifts the trunk's representation quality asymptotically.** REPA variants open a quality gap over baseline that never closes. The random control gains least, evidence that a trained encoder contributes a genuine mechanistic effect rather than mere regularisation from the auxiliary loss. (See Appendix A.3 for the probe protocol and further ablations.)

ation quality. We now turn to the results.

6.2 Results

6.2.1 REPA improves representation quality asymptotically

At baseline, the flow-matching objective helps the model encode some useful representational information. Figure 6.2 shows the results of probing our models along the three orthogonal axes of §6.1.3: local chemical environment (inverse folding), local geometry (backbone dihedrals), and global fold (CATH classification, with CATH-A as our headline metric). The trained baseline consistently does better than both random features and an untrained Proteina trunk. This suggests that the flow-matching objective does indeed learn to *represent* protein backbones as it learns to *generate* them.

Every trained-encoder REPA variant improves representational capacity, consistently and durably. Each lifts all three probes substantially above the baseline, from the very beginning of training, and the difference persists. This is an asymptotic gap that the baseline never closes across 1.8M steps, not just a head start or an accelerated convergence pattern. This gap confirms one of the hypotheses posited in the original REPA presentation [5]: an alignment target during training can fill the representational headroom left by the flow-matching objective.

Every encoder routes representation gain differently. REPA-GearNet consistently wins the fold probes, while REPA-MPNN wins the per-residue probes (Table 6.2). This pattern follows what we would expect a priori, given the tasks that each encoder has been specifically trained for. This ties into our emerging theme on the multi-factorial nature of representation learning in the molecular domain (Chapter 3). No single encoder captures every factor, so the choice of alignment target is also a choice over the representational

Variant	IF top-1 ↑	dihedral MAE (°) ↓	CATH top-1 acc. ↑		
			C	A	T
Baseline	0.130	27.6	0.717	0.397	0.303
REPA L4-random	−0.002	+2.7	+0.051	+0.029	+0.017
REPA L9-GearNet	+0.022	−3.6	+0.163	+0.283	+0.462
REPA L9-MPNN	+0.027	−6.8	+0.085	+0.078	+0.113

Table 6.2: **REPA offers persistent and encoder-routed representation advantages.** Consistent with what each encoder was trained for, REPA-GearNet wins the fold probes (CATH C/A/T), while REPA-MPNN wins the per-residue probes (IF, dihedral). We headline on CATH-A, and report C and T for completeness. (Single-seed; averaged over the 700K–1.2M training-step window.)

factor a model designer wishes to emphasise. We establish this routing on PDB-trained models and report the AFDB results in Appendix A.5.

REPA offers a mechanistic effect beyond just regularisation. Our random-weights control improves on the baseline for the fold probes (CATH), but not for the per-residue ones. Where it helps, this may be because *any* regularisation generically improves representation capacity, or because even an untrained message-passing network imposes some geometric priors that aid the trunk. However, the trained encoders consistently perform far better. This sharpens the case for a genuine mechanistic effect, as regularisation alone is not the whole story.

The random control behaves as we expected a saturated target would. In §4.3 we characterised the random-init encoder as saturated, based on its high-floor signature. This makes the alignment loss cheatable, in that the trunk can optimise the loss without learning anything useful (§4.1). Our results bear this out. The random encoder's cosine similarity saturates higher than every trained model's — ≈0.89 against ≈0.80 for a trained GearNet at the same layer 4 on PDB. Yet this near-perfect alignment buys the least representation quality of any variant, and no FPSD or designability gain (§6.2.3).

6.2.2 A diagnostic check on alignment

Following Yu et al. [5] and recent in-domain work [8], we investigate the mechanism that may be driving REPA's representation quality gains by quantifying the representation alignment it induces. Using CKNNA [65], we score how closely our models' per-residue geometry matches that of three domain-specific encoders: GearNet and ProteinMPNN, our two alignment targets in this chapter; and ESM2 [24], which we align to only in our ablations, not in the models presented here. We note that we study alignment only as a diagnostic, and not to make claims about REPA's utility.

REPA lifts alignment above the random floor, albeit at low magnitude. The baseline sits at the random floor at every layer (CKNNA ≈ 0), while REPA separates from it cleanly, suggesting that its hidden states do move towards the encoder's (Figure 6.3a). This serves as a minimal check that the construction changes model behaviour. However,

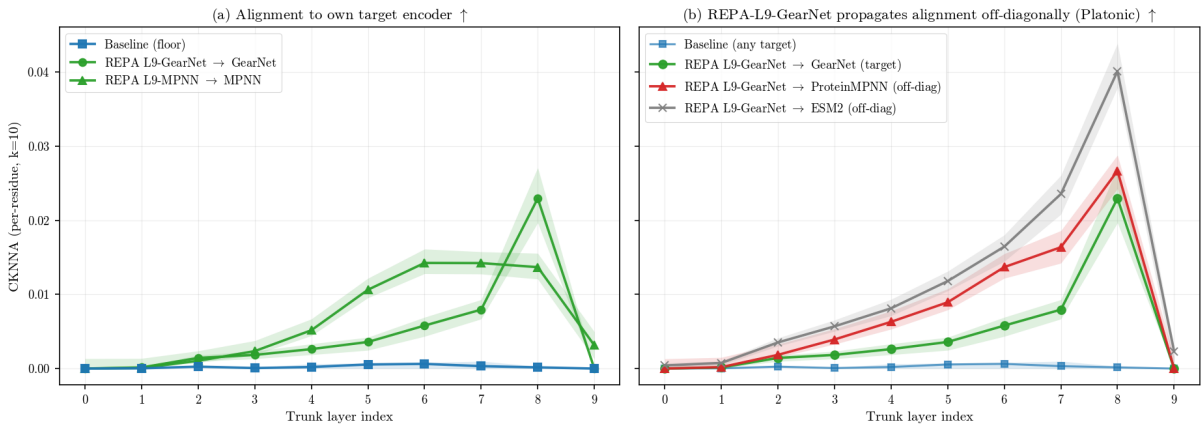

Figure 6.3: **REPA aligns the trunk with domain encoders, including those it was not trained with.** (a) Each REPA variant lifts per-residue alignment to its own target encoder above the baseline floor. (b) REPA induces cross-encoder alignment: for example, a GearNet-aligned trunk also moves toward ProteinMPNN and ESM2. Absolute alignment is small throughout, so we read the pattern, not its magnitude. (Step 1M, t=1.0; per-residue CKNNA, k=10; lines are bootstrap medians, shaded bands the 5–95% subsample bootstrap interval.)

the magnitude of this signal is small. Peak per-residue CKNNA (bootstrap median) is $\approx$ 0.02–0.05, on a scale where self-alignment is 1. We therefore read the pattern, not its size, and stake no strong claim on these numbers.

Aligning to one encoder nudges the trunk towards others as well. Every REPA variant we tested raises alignment to all three encoders, including those it was not aligned to (full model×encoder matrix in Appendix A.6). A GearNet-aligned trunk, for example, also moves closer to ProteinMPNN and ESM2 [24] (Figure 6.3b), suggesting that REPA-induced alignment is not merely circular self-alignment. One possible explanation for this phenomenon is the *Platonic representation hypothesis* [65]. This idea suggests models trained on related data converge toward a shared representation. In this framework, REPA may be nudging the trunk toward a common structural representation that several encoders approximate, rather than toward any single encoder.

6.2.3 REPA accelerates generation quality by climbing the representation–generation envelope faster

Here, we outline our headline results using the *Quality* metrics of our suite, FPSD and designability (as in Table 3.2). A deeper analysis of generation quality along tertiary- and secondary-structure decomposition follows in §6.2.5.

REPA gives generation quality an early head start. At 400K steps, every variant already leads the baseline on PDB designability, by +48% to +149%. Most variants also lead on FPSD in both regimes, with the best at +28%. AFDB designability is the lone exception, for the proxy reason given in §6.1.1 (Table 6.3).

This acceleration is consistent with the representation-bottleneck hypothesis.

Variant (last ckpt)	**Accel.** @400K (vs base)	**Long run** (abs. best)
AFDB FPSD ↓		
baseline (to 1.8M)	431 (@400K)	386 (@1.3M)
L4-GearNet (1.3M)	+24% (328)	282 (@1.2M)
L9-GearNet (900K)	+28% (313)	313 (@400K)
L4-MPNN (1.1M)	−10% (477)	416 (@1.0M)
L9-MPNN (1.5M)	+18% (352)	352 (@400K)
Random control (500K)	+16% (361)	361 (@400K)
AFDB designability ↑		
baseline (to 1.8M)	0.75 (@400K)	0.75 (@400K)
L4-GearNet (1.3M)	+4% (0.78)	0.78 (@400K)
L9-GearNet (900K)	−4% (0.72)	0.77 (@800K)
L4-MPNN (1.1M)	−9% (0.68)	0.69 (@700K)
L9-MPNN (1.5M)	−6% (0.71)	0.77 (@700K)
Random control (500K)	+5% (0.79)	0.85 (@500K)
PDB FPSD ↓		
baseline (to 2.4M)	472 (@400K)	288 (@1.8M)
L4-GearNet (1.0M)	+7% (440)	432 (@1.0M)
L9-GearNet (1.6M)	+28% (341)	319 (@700K)
L4-MPNN (1.6M)	−8% (512)	329 (@1.6M)
L9-MPNN (2.0M)	+13% (410)	297 (@1.6M)
Random control (1.2M)	−7% (506)	351 (@1.1M)
PDB designability ↑		
baseline (to 2.4M)	0.26 (@400K)	0.67 (@1.6M)
L4-GearNet (1.0M)	+98% (0.52)	0.70 (@700K)
L9-GearNet (1.6M)	+97% (0.52)	0.63 (@800K)
L4-MPNN (1.6M)	+124% (0.59)	0.71 (@1.6M)
L9-MPNN (2.0M)	+149% (0.66)	0.82 (@1.8M)
Random control (1.2M)	+48% (0.39)	0.46 (@900K)

Table 6.3: **REPA accelerates generation quality over the baseline.** In some regimes it also wins the long run. Acceleration is the %-delta vs the baseline at 400K (the anchor of Figure 6.4), with the absolute score in parentheses; the long run is each variant's absolute best within a common 2.0M-step window, coloured against the baseline-best row.

Yu et al. [5] hypothesise that a model's representation quality and its generation quality are coupled, because it is learning both at once. In our PDB-trained models, this coupling is clear. Across checkpoints, representation and generation quality are positively correlated, and the strongest of these correlations survive controlling for training step (Table 6.4). The same coupling holds on AFDB-trained models, and in the same encoder-matched form (Appendix A.7). If the two move together, then securing good representations early should secure good generation early as well.

REPA climbs the representation–generation envelope faster than the baseline. REPA reaches an asymptotically higher representation quality almost as soon as training begins (§6.2.1, Figure 6.2). On the representation–generation curve posited by the bottleneck hypothesis, this places it further along than the baseline at matched compute (Figure 6.4, shown for the strongest PDB variant). Hence, it reaches a given level of

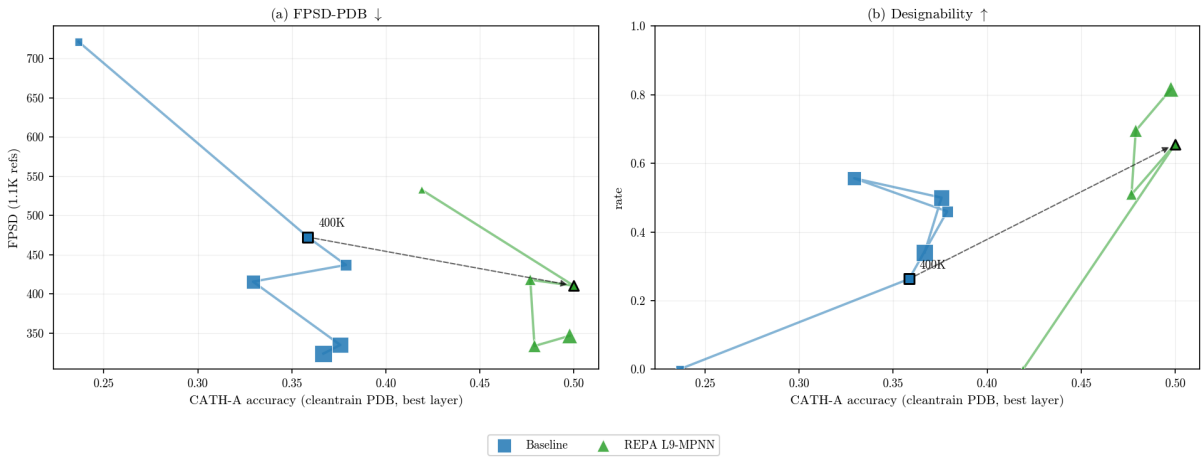

Figure 6.4: **At equal compute, REPA achieves better representation and generation quality.** The dashed arrow marks the matched-compute comparison at step 400K. Against the baseline, the REPA variant has higher fold accuracy (CATH-A) and better generation: lower FPSD (a) and higher designability (b). (Markers are a coarse ≈400K-spaced subset of checkpoints from 100K, with the 400K point bordered; x = student CATH-A accuracy, higher to the right.)

Representation quality	**FPSD**		**Designability**	
	partial	raw	partial	raw
Inverse folding (top-1)	+0.37	+0.17	+0.79	+0.61
Dihedral MAE	+0.44	-0.03	+0.77	+0.36
CATH-A	+0.22	+0.11	+0.49	+0.38

Table 6.4: **Better trunk representations track better generation — strongly for designability, moderately for FPSD.** Partial Pearson correlation controlling for training step, with the raw value alongside; oriented so a positive value means better representation accompanies better generation. (n=59 checkpoints pooled over the baseline, the REPA variants, and the random control.)

generation quality at less training. We interpret REPA as simply traversing this curve faster, rather than finding a new shortcut.

This representation–generation link is encoder-matched, suggesting the effect is encoder-specific, not generic regularisation. Each encoder's generation gain lands on the axis where its representations are strongest. GearNet, the better fold encoder (§6.2.1), gives the larger gain in fold *distribution* (Table 6.7). MPNN, the better local encoder, gives the larger acceleration in *designability* (Table 6.3). Such axis-matching would be unlikely under a generic auxiliary loss, but we do not claim a formal proof of causality.

Broadly, REPA's lasting advantage tracks the headroom the baseline leaves. Where the baseline plateaus poorly, REPA's lead is large and permanent, as on AFDB FPSD. Where the baseline is a slow learner, the lead narrows yet still holds, as on PDB designability. Only where the baseline is a genuinely strong learner, as on PDB FPSD, does it eventually close the gap. AFDB designability, again, leaves little headroom over

	Sampler setting					
Metric (700K)	ODE	γ=0	γ=0.35	γ=0.45	γ=0.5	γ=1.0
PDB-trained — baseline vs. REPA L4-MPNN						
FPSD↓, base	353	782	462	437	425	508
FPSD↓, REPA	279	561	371	349	350	424
Des↑, base	0.04	0.84	0.56	0.46	0.36	0.01
Des↑, REPA	0.16	0.78	0.66	0.64	0.52	0.11
AFDB-trained — baseline vs. REPA L4-GearNet						
FPSD↓, base	220	641	522	494	475	366
FPSD↓, REPA	148	586	328	298	292	276
Des↑, base	0.12	0.91	0.72	0.72	0.69	0.21
Des↑, REPA	0.35	0.96	0.84	0.77	0.72	0.28

Table 6.5: **At almost every sampler setting, REPA reaches a higher level of both FPSD and designability than the baseline.** (Absolute values at step 700K; the REPA row is coloured against the baseline. Baseline vs. a representative REPA variant per dataset; FPSD is whole-set distribution match, Des is designability; γ=0.45 is Proteina's default. The lone exception is PDB designability at the degenerate γ=0 setting, where the baseline saturates by repeating a few folds.)

the baseline. However, as we noted earlier (§6.1.1), this is likely a quirk of the relationship between how designability is measured and how AFDB was generated, and not due to any special merit. Tellingly, even the random control scores highly here.

6.2.4 REPA's gains are robust to sampler noise

The FPSD and designability gains we observe hold across a sampler noise ablation. At almost every γ setting, REPA reaches a higher level of *both* FPSD and designability than the baseline (Table 6.5). The exception is the degenerate γ=0 regime, where the baseline saturates designability by repeating a few folds. This suggests that REPA's gains are built into the learned model, not merely recovered at inference.

REPA raises the deterministic-sampling floor, the strongest sign of a better velocity field. The probability-flow ODE samples the trunk's learned distribution without injected noise (§2.1), the closest view of the true learned distribution before any correction. Under the baseline, that distribution is barely designable: Des 0.04 on PDB and 0.12 on AFDB. But REPA's learned distribution is already far more designable at ODE, across every learned encoder (Table 6.6). This suggests REPA mechanistically improves the distribution the trunk learns.

Having established REPA's advantages on our headline metrics, we now seek to understand the nuances of what the models learn.

Variant (700K)	ODE designability
PDB-trained	*(baseline = absolute; REPA = Δ vs. baseline)*
Baseline	0.04
REPA L4-random	−0.02
REPA L4-GearNet	+0.16
REPA L9-GearNet	+0.04
REPA L4-MPNN	+0.12
REPA L9-MPNN †	+0.15
AFDB-trained	
Baseline	0.12
REPA L4-GearNet	+0.23
REPA L9-GearNet	+0.21
REPA L9-MPNN	+0.10

Table 6.6: **Every REPA-learned model yields more designable proteins than baseline when sampled deterministically.** The ODE limit samples the trunk's learned distribution directly, which suggests REPA learns a better underlying distribution rather than merely benefiting from sampler noise. (ODE designability at step 700K; $|\Delta|<0.025$ uncoloured. Single-seed. †PDB L9-MPNN has no 700K ODE data, so it is shown at 1.0M, with its Δ taken against the baseline at 1.0M.)

6.2.5 REPA consistently dominates baseline on whole-set, but not on designable-subset metrics

Two facts sit side by side here, and they are not in tension. Over the whole generated set, REPA genuinely improves the distribution: more diverse, and closer to the reference. But within the *designable* subset, it may only be moving along the designability–diversity envelope (§3.6), trading some of that subset's diversity for the designability it gains.

REPA consistently improves whole-set distribution match and diversity, and the gain is durable. Under almost every REPA variant tested, the generated set matches the reference fold distribution more closely (fJSD-A) and spreads more widely across fold classes (fS-A); secondary-structure distribution match (ssJSD-2D) improves likewise. Curiously, this holds for the random-weights control too, but only the learned encoders translate it into higher FPSD and designability (Table 6.7). The advantage is not a transient head start. The whole-set lead persists deep into training, with the strongest variant on each dataset still ahead at 1.0M steps — REPA-L9-MPNN on PDB (FPSD −81, fold-score +0.29) and REPA-L4-GearNet on AFDB (−228, +3.68) — even as its designable-subset diversity stays below baseline.

However, REPA's designable subset displays fold concentration. Despite the whole-set gains, within the designable subset, REPA's pwTM rises above the baseline at almost every checkpoint, and sharply so by 1.0M steps (Table 6.8).

The concentration is encoder-dependent and severe for β-rich folds. In our data, the concentration appears especially severe for β-strand-rich samples (Table 6.9). The REPA construction appears to generate the *same* β-rich folds repeatedly, while the

	Quality		T-W		T-D		S-W	S-D	
Variant (700K)	Des↑	FPSD↓	fJSD-A↓	fS-A↑	#Clust↑	pwTM↓	ssJSD2D↓	ssJSD2D↓	E%des↑
PDB-trained, n≤256, 700K, 3 seeds; FPSD is FPSD-PDB (in-distribution).									
Baseline	0.46	437	1.11	5.48	77	0.27	0.35	0.40	0.22
REPA L4-random	−0.08	+34	−0.55	+0.05	−23	−0.10	−0.13	−0.01	−0.11
REPA L4-GearNet	+0.24	+71	−0.38	+1.63	+1	+0.11	−0.21	−0.12	+0.00
REPA L9-GearNet	+0.14	−118	−0.57	+2.31	+27	−0.02	−0.19	−0.10	−0.04
REPA L4-MPNN	+0.18	−88	−0.13	+0.48	−4	+0.08	−0.16	−0.14	+0.01
REPA L9-MPNN	+0.05	−19	−0.32	+0.37	+33	−0.01	−0.14	−0.11	−0.07
AFDB-trained, n≤256, 700K, 2–3 seeds (L4-MPNN: 1 seed); FPSD is FPSD-AFDB.									
Baseline	0.72	494	2.45	4.59	128	0.22	0.15	0.16	0.15
REPA L4-GearNet	+0.05	−195	−1.56	+2.27	−35	+0.04	−0.11	−0.07	−0.02
REPA L9-GearNet	+0.01	−172	−1.29	+2.76	−50	+0.11	−0.12	−0.06	+0.04
REPA L4-MPNN	−0.03	−22	−0.39	+0.03	−1	−0.07	−0.02	+0.05	−0.01
REPA L9-MPNN	+0.05	−139	−0.93	+0.31	+21	−0.03	+0.01	+0.02	−0.04

Table 6.7: **Across the metric suite, almost every REPA variant improves whole-set distribution match and diversity over baseline.** The effect is broad enough that even the random-weights control matches the reference distribution better. However, only the learned encoders also lift per-sample quality (FPSD and designability). (Step 700K; header arrows mark the good direction; whole-set diversity is read from fS-A, designable diversity from #Clust and pwTM.)

	Baseline		REPA L9-GearNet		REPA L9-MPNN	
Step	whole	des.	whole	des.	whole	des.
400K	0.137	0.143	0.159	0.158	0.164	0.172
700K	0.221	0.229	0.189	0.280	0.215	0.209
1.0M	0.173	0.168	0.235	0.491	0.241	0.342

Table 6.8: **REPA's whole set stays as diverse as the baseline, but its designable subset is more concentrated.** Mean pairwise TM-score (pwTM; lower is more diverse) over the whole generated set and the designable subset. pwTM is an average, so it compares across models and sample counts. Whole-set pwTM tracks the baseline, but designable pwTM rises above it at almost every step (red = REPA more concentrated than baseline; green = more diverse). The concentration is specific to the designable subset, not the whole distribution. (PDB-trained, $n \leq 256$, last-layer GearNet and ProteinMPNN.)

baseline spans many. Notably, the AFDB-MPNN variant does not display this: its designable subset stays as diverse as the baseline's in every β-bin (Table 6.9). One explanation is that encoder routing leads to different secondary- and tertiary-structure distributions; this MPNN configuration nudges the learned distribution towards helices, which may be why it escapes this largely strand-heavy phenomenon (Appendix A.9).

On PDB, even the baseline concentrates early in training, but grows out of it. After 700K training steps, both the PDB baseline and its REPA variants concentrate their β-rich designable samples, though REPA already does so more. By 1.0M steps, the baseline has diversified while REPA stays concentrated, widening the gap (Table 6.9). This suggests REPA does not *engender* the concentration so much as fail to grow out of an early-training regime the baseline leaves behind. We read the late checkpoints as a

Designable pwTM ↓	**700K steps**			**1.0M steps**		
Variant	β<10	10–25	β≥25	β<10	10–25	β≥25
PDB-trained						
Baseline	0.15	0.22	0.50	0.15	0.23	0.13
REPA L9-GearNet	0.16	0.27	0.73	0.14	0.24	0.87
REPA L9-MPNN	0.16	0.21	0.67	0.26	0.22	0.67
AFDB-trained						
Baseline	0.14	0.29	0.17	0.16	0.26	0.15
REPA L9-GearNet	0.37	0.31	0.82	0.40*	0.32*	0.76*
REPA L9-MPNN *(falsifier)*	0.15	0.27	0.11	0.15	0.24	0.15

Table 6.9: **A REPA-learned distribution may collapse onto fewer distinct folds within its designable subset.** The effect is strongest in the β-strand-rich folds. (Designable-subset pairwise TM, lower = more diverse; columns bin the designable subset by β-strand fraction (%); here, unlike elsewhere, colour marks REPA *vs.* baseline at the same step and bin: red = more concentrated, green = more diverse, $|\Delta|<0.05$ uncoloured. Single-seed; *900K steps.)

single-seed directional pattern, not a precise effect size.

Does alignment outlive its purpose? Given REPA's early acceleration in generation quality and the later-stage fold concentration we observe, the best approach may be to stop alignment after early training. A similar arc is reported in the image-diffusion literature, where REPA helps early and is best early-stopped [13]. We do not test such a schedule and make no argument of this form, but on this reading an early stop may let REPA diversify like the baseline while keeping the whole-set gains.

6.2.6 What we do not claim

Several tempting claims do not survive our data, and we flag them here.

- *No asymptotic generation quality win on experimental data.* REPA accelerates PDB convergence, but the lead is transient. The baseline matches REPA's FPSD plateau by ∼1.4M steps and edges below it by 1.6M (§6.2.3). The durable advantage on distribution match is on synthetic (AFDB) data, where the baseline plateaus poorly.
- *Unclear designability gains on synthetic data.* At the recommended sampler setting, AFDB designability is already biased upwards by construction and near-saturated for the baseline, so a reliable whole-set gain is hard to establish (§6.1.1). The deterministic ODE floor of §6.2.5 is the exception.
- *No reliable novelty gain.* The signal is too noisy to call in either direction.
- *No general "β-preservation".* The secondary-structure shift is encoder×dataset conditional, and MPNN-AFDB reverses it (§6.2.5).
- *Co-movement, not mediation.* Representation and generation quality move together. We do not claim one formally causes the other, but we note that the pattern is consistent with the representation bottleneck hypothesis (§6.2.3).

- *Unconditional generation only.* Every experiment here is unconditional: we specify a target length but never a fold class or other structural motif. We do not claim these dynamics carry over to the conditional setting (§2.1).
- *No claim of optimality.* We do not search the configuration space for the best setup of layer, weight, or encoder. The question here is whether the mechanism works at all, not how to tune it for maximum gain (§6.1).

6.3 Discussion

We show that REPA accelerates Proteina's convergence on generation quality, and suggest that this works through representation. Every trained encoder teaches the trunk to represent structural characteristics that the flow-matching loss never learns on its own. This representation gain is mirrored in the generation gain, encoder by encoder, on the axis each encoder is strongest. A random-weights control suggests that the mechanism is indeed genuine structural transfer, not the side effect of an auxiliary loss. Its alignment loss drops more than any trained encoder, yet this buys the least representation quality, and no generation gain at all (§6.2.1).

It is likely that REPA improves the distribution learned by the model. Two observations point this way. The representation gap opens early and never closes (§6.2.1), and the generation advantage holds for the learned encoders even in the deterministic ODE limit, which reads the learned distribution with no injected noise (§6.2.5). The REPA advantage also holds across all sampler noise ablations, suggesting again that the gain is built into the model, rather than recovered at inference.

The same encoder pull that broadens the whole set may be what keeps the designable set concentrated. Relative to baseline, REPA broadens the folds the model generates over the whole set, but may concentrate the designable subset onto fewer modes. This concentration appears early in the baseline as well as in REPA. The two then part ways: the baseline grows out of the early regime and diversifies, while REPA stays concentrated, so the cost relative to baseline widens deep into training rather than appearing there for the first time. One interpretation may be that the encoder is a fixed target, so once the trunk has extracted the representational signal it had to offer, continued alignment keeps pulling toward the encoder's own distribution, which for our structure encoders is comparatively narrow. In this framing, the gain and the cost are not separate phenomena, but early and late phases of one pull. Indeed, the image-diffusion literature reports a similar arc, and early-stops alignment for it [13]. Nevertheless, in this report, we do not investigate the mechanism behind the data we observe, and we do not claim early-stopping would help here.

REPA helps most where the baseline struggles most. On experimental data, the FPSD acceleration is real but transient, and the baseline eventually catches up. On synthetic data, the FPSD baseline plateaus poorly, and the REPA lead is durable. The

intervention is identical in both cases, and the asymmetry tracks the baseline's capacity.

REPA is a family of encoder-mediated interventions, not a single mechanism. Our results suggest that in a domain with multi-factorial representation, choosing an alignment target implicitly chooses what factor gets boosted. There is no single best encoder. Each one buys a different representational factor, so the optimal outcome is likely use-case specific.

We carry these results, and the contrast with the small-molecule study of Chapter 5, into this report's conclusion in Chapter 7.

Chapter 7

Conclusions

Our work is motivated by a tension in the field of molecular generation, as outlined in Chapter 1. Following a wider trend in machine learning, molecular generation has traded hard structural priors for architectural simplicity, and curated experimental data for synthetic scale. Both phenomena dilute the structural signal available to models during training, which exacerbates their representation bottleneck [5] and causes poor convergence behaviour. Representation alignment (REPA) is potentially one cheap way to provide models this signal and accelerate training. This report investigates two questions: does REPA work in the molecular domain, and when does it work at all?

In this concluding chapter, we collect threads from across our report to build answers. We also raise new questions and future work that follow from our report, and invite a larger discussion in this field.

7.1 Does REPA work in the molecular domain?

Our report documents two regimes, small molecules and protein backbones, and presents a different answer for each. On small molecules, no REPA variant left a measurable mark on Tabasco (Chapter 5). On protein backbones, REPA accelerated Proteina's convergence on the two key measures of generation quality, likely by improving the representations learned by the model (Chapter 6).

What explains these divergent behaviours? At root, the two regimes represent completely different chemistry. A small drug-like molecule and a folded protein backbone are not the same kind of object, even if their in-silico parameterisations appear to squash these distinctions at first glance. That chemistry sets the combinatorial space a generator must cover: modest and topology-dominated for small molecules, vast and hierarchical for proteins. The former presents a more learnable target, so its generators mature sooner. Indeed, by the time we test REPA, the evaluation metrics for small-molecule generation sit at ceiling, and the baseline Tabasco model already learns the information contained in our external encoders. Protein generation, working in a far larger space, is far from this point, and therefore benefits from the intervention.

The protein-backbone result is ultimately a statement about training efficiency, and it pays off along two axes.

The first is *acceleration*: for a fixed compute budget, REPA reaches better generation quality in both the experimental (PDB) and synthetic (AFDB) regimes. Early in training, every variant already leads the baseline on designability, and most lead on distribution match (FPSD) in both regimes (§6.2.3). A cheap auxiliary signal buys faster convergence, which is highly valuable in a data-scarce, prior-light field.

The second axis is *asymptotics*: whether that early lead survives all the way to convergence. Here the answer is not uniform, and it tracks the headroom the baseline itself leaves rather than anything REPA does. Where the baseline plateaus poorly, the lead is permanent, as on AFDB distribution match; where the baseline is a genuinely strong learner, it eventually closes, as on PDB distribution match (§6.2.3).

Both axes point the same way as a data-efficiency story. REPA always buys a faster climb, and can offer a lasting advantage when the baseline struggles most. The field's frontier — larger synthetic corpora, longer backbones, the all-atom setting — is where the generation problem is hardest and an unaided baseline has ground to make up. This may also be where REPA's payoff is greatest, and where it should be tested next.

7.2 When does REPA work?

The discussion above already hints at our answer, which is that REPA works when two conditions hold together. The evaluation surface must leave *headroom* for a gain to register, and the encoder must be a *suitable teacher* — it must carry structural information the generator does not already learn cheaply.

In this framework, our small-molecules study showed little promise because generation and representation quality metrics were already saturated, and the trunk already learned everything our encoders carried. Indeed, the problem of generating structurally plausible small molecules in-silico appears largely solved. To set appropriate expectations about what this means, we note that structural plausibility is only one desideratum among the many a real design campaign cares about (§2.3), and it is arguably the most tractable of them. The properties that actually determine a molecule's utility — how it binds a target, metabolises, or synthesises — are downstream of plausibility and far less settled. Moreover, all of this is measured in an in-silico sandbox that is itself a coarse stand-in for real chemistry. The saturated metrics do not suggest that we have mastered small-molecule generation, but that one tractable sub-problem of it looks solved under proxies that are far removed from the wet lab. But this suffices to leave REPA nothing to add.

Protein backbone generation presents a more challenging task, where both representation and generation metrics are unsaturated, and determining teacher suitability remains difficult. In Chapter 3, we propose a new framework in which to interpret the vast array of relevant metrics, which are complex both in isolation and in their relationships with each

other. We also frame the central dynamic that characterises a generator's quality, the *designability–diversity* trade-off, and the distinction between merely moving along such an envelope and pushing it outward. In Chapter 4, we then propose a set of heuristics to estimate the utility of an encoder as a REPA teacher. We investigate the *information* content of a suite of pre-trained encoders, the *relevance* of that content to the generative task, and the mathematical *conditioning* of their embeddings, and use the most promising candidates in the studies that follow. We note that the approach we propose here is domain-agnostic. The questions we raise are purely statistical, and can be asked of any encoder in any domain with a few modifications.

However, the molecular domain presents unique challenges in how multi-factorial and multi-scale its representations are, especially in contrast to images (§3.2). In this light, our work also presents a new view on REPA: it is not a single mechanism, but rather a *family of interventions*. Every pre-trained encoder emphasises a different factor, so the choice of alignment target is also a choice over the structural factor one seeks to enhance.

7.3 Open questions

We close by looking ahead, and proposing new directions based on our work.

Does REPA work in other scientific domains? Our report opened by placing this work in the wider arc of generative machine learning for the sciences (Chapter 1). The same paradigm of prior-light models, sparse data, and multi-scale representation recurs across the sciences. For example, in fluid mechanics, diffusion models are learning to generate turbulent flow fields [22]. Alleviating the representation bottleneck could open the door for more efficient and performant models here, following our work with proteins.

Does our encoder profiling framework generalise? Singh et al. [6] ask the which-encoder question for vision, and answer it with structural metrics tied to the image grid. We attempt to generalise this analysis into a set of domain-agnostic heuristics. An open question is whether these are two views of a single, canonical characterisation of what makes a good alignment target. We propose retrospectively analysing published REPA work across domains, to see if the framework offers useful guidance.

We read our contribution not as a verdict that REPA accelerates protein backbone generation, but as a principled framework for assessing how to improve generative models for science more broadly. In the context of that aspiration, we hope to raise more interesting research questions than we answer, and to push the evolution of this exciting space.

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Appendix A

Technical details

A.1 Dataset composition and split overlap

Table A.1 summarises the two datasets used in Chapter 6. Both training sets are α/β-dominated and span a substantial fraction of CATH architectures (44.2% PDB, 61.8% AFDB), which is what makes the per-chain CATH fold probe meaningful. Neither random split is sequence-clustered: 98.3% of PDB and 92.2% of AFDB validation chains have a $\geq$30%-identity homolog in their own training set, so absolute in-house scores are inflated and we read rank-order and Δ-from-baseline instead. The cross-database overlap is far lower — only 62.1% of AFDB validation chains have a $\geq$30%-identity hit in PDB training — so a homology-filtered AFDB set serves as a cleaner probe set for PDB-trained models (Chapter 3).

A.2 Choice of CATH fold-probe level

We report CATH *architecture* (CATH-A) as the headline fold-probe level throughout Chapter 6. Table A.2 justifies that choice: Class is too coarse (four buckets, one already near-saturated), Topology too finely-bucketed to probe reliably, and Architecture sits between — discriminative but learnable.

Property	**PDB**	**AFDB (Swiss-Prot)**
Origin	Experimental (X-ray, cryo-EM)	Predicted (AlphaFold2, v6)
Residue-length crop n	$\leq$ 256 residues	
Train / val / test chains	273,771 / 3,190 / 323	229,670 / 4,521 / 235
CATH coverage (train)	44.2%	61.8%
Val$\leftrightarrow$train, $\geq$30% id.	98.3%	92.2%
AFDB-val$\leftrightarrow$PDB-train, $\geq$30% id.	62.1%	

Table A.1: Properties of the two datasets we study, at the headline $n\leq$256 regime. Sequence identity is computed with MMseqs2 [29] (`easy-search`, $\geq$30% identity over $\geq$80% alignment coverage).

CATH level	#classes	baseline probe acc.	verdict
C (class)	4	0.733	one class (α/β) is 55% of the data; coarse, already near-saturated — *gameable*
A (architecture)	∼40	0.407	balanced and discriminative, with headroom to move — **our headline fold metric**
T (topology)	∼1400	0.301	long-tailed (top fold ∼18%; only ∼89 classes populated enough to probe); per-class accuracy is unreliable — *too many buckets*

Table A.2: **We read fold-level representation quality off CATH *architecture* — Class is too coarse to discriminate, Topology too finely-bucketed to probe reliably.** Class has 4 buckets (one at 55%, baseline already 0.73); Topology has ∼1400 imbalanced buckets; Architecture sits between — enough classes to be meaningful, enough examples per class to be learnable — so CATH-A is our headline fold metric (C and T are reported in the appendix). ("Baseline probe acc." is a linear probe on the un-aligned trunk.)

A.3 Leakage-controlled probe evaluation

The random train/val split (§A.1) is not sequence-clustered, so probe scores are exposed to two distinct leakages. *Model leakage* is asymmetric: a model trained on the leaky split has effectively seen near-duplicates of the evaluation proteins, an advantage a differently-curated model (such as an externally-pretrained checkpoint) does not share. *Probe leakage* is symmetric: a probe fit on training chains near-identical to its evaluation set can shortcut by matching sequences rather than learning generalisable structure, inflating every model's absolute score.

We control these by filtering the evaluation. A *probe-clean* split removes probe leakage alone — the probe's training chains are filtered to have no ≥30% sequence-identity hit to the evaluation set, which is otherwise the full validation set. A *blinded* split removes both — the evaluation proteins are held to under 30% identity to all training data (here, a cross-database set of AFDB chains with no such hit in either the PDB or AFDB training pool, ∼325 proteins). We use the blinded split for the per-residue probes (inverse folding, dihedral), where the ∼325 proteins still yield ∼43k residues — full statistical power. The per-chain CATH probe is different: fold classification needs many distinct chains spread across populated fold classes, and the blinded slice is too small and too class-sparse for this, so we relax it to the probe-clean split on the fuller validation set (∼3,190 chains). Comparing the splits also decomposes the leakage — the dirty-to-probe-clean drop isolates probe leakage and the probe-clean-to-blinded drop isolates model leakage — and the REPA-over-baseline gap survives in every regime.

Variant	IF top-1 ↑	dihedral MAE (°) ↓	CATH top-1 acc. ↑		
			C	A	T
Baseline	0.130	27.6	0.717	0.397	0.303
REPA L4-random	−0.002	+2.7	+0.051	+0.029	+0.017
REPA L4-GearNet	+0.015	−6.1	+0.151	+0.212	+0.304
REPA L4-MPNN	+0.032	−3.5	+0.109	+0.128	+0.168
REPA L9-GearNet	+0.022	−3.6	+0.163	+0.283	+0.462
REPA L9-MPNN	+0.027	−6.8	+0.085	+0.078	+0.113

Table A.3: **Full PDB representation-quality ablation across all six variants.** The remaining encoder×depth combinations behind the four-row main-text Table 6.2. The encoder-routed pattern holds throughout: ProteinMPNN wins the per-residue probes (IF, dihedral), GearNet wins the fold probes (CATH C/A/T). At matched layer 4, the trained GearNet (L4-GearNet) beats the random control (L4-random) on every probe, so the learned-versus-random gap is not an artefact of injection depth. (Baseline absolute; REPA rows are Δ-from-baseline. Best-layer linear probe, n_{train}=1000, seed 42, mean over the 700K–1.2M window; IF/dihedral on the cross-database blinded set, CATH on the cleantrain set. green / darker = improvement/best per probe. In-window step coverage is uneven across rows.)

A.4 Full representation-quality ablation

The main-text representation table (Table 6.2) shows four rows. Table A.3 gives the full six-variant grid, adding the layer-4 GearNet and ProteinMPNN combinations. Two readings carry over from the main text: the encoder-routed pattern (ProteinMPNN wins the per-residue probes, GearNet the fold probes) holds across all depths, and at matched layer 4 the trained GearNet beats the random control on every probe.

A.5 Representation quality on AFDB

Table A.4 is the AFDB-trained counterpart of the PDB Table 6.2. The fold routing (GearNet → CATH) replicates; the per-residue routing is weaker and noisier than on PDB, for the reasons given in the caption.

A.6 Representation-alignment matrix

The alignment diagnostic in §6.2.2 reports peak per-residue CKNNA for a single illustrative variant. Table A.5 gives the full model×encoder matrix behind that reading. It confirms the off-diagonal generalisation we describe there. Aligning the trunk to one encoder raises its CKNNA to the other two as well, not just the alignment target. The peak sits at the last trunk layers (L7–L8 here). The absolute values stay an order of magnitude below image-domain REPA, so we read the direction and layer pattern rather than the raw magnitude.

Variant	IF top-1 ↑	dihedral MAE (°) ↓	CATH top-1 acc. ↑		
			C	A	T
Baseline	0.126	35.7	0.767	0.467	0.250
REPA L4-GearNet	−0.004	+5.1	+0.100	+0.089	+0.083
REPA L9-GearNet	−0.001	+1.5	+0.083	+0.067	+0.167
REPA L4-MPNN	+0.000	+13.7	+0.011	+0.044	+0.194
REPA L9-MPNN	+0.001	+4.3	+0.033	−0.067	+0.250

Table A.4: **On AFDB, REPA's representation gain concentrates on fold structure.** The AFDB counterpart to Table 6.2. REPA-GearNet lifts the CATH fold probes here as on PDB, so the fold routing replicates. The per-residue picture does not: inverse-folding accuracy is flat, and the dihedral probe shows no consistent REPA effect (it tracks a baseline that itself swings several degrees between checkpoints), unlike the clear MPNN-led per-residue gain on PDB. At this checkpoint density the per-residue reads are noise-dominated, so we lean on the fold result. ($n \leq 256$ AFDB-trained; best-layer linear probe at $n_{\text{train}}=1000$ on the cross-database blinded set; mean over the 700K–1.2M window; Δ from baseline; green / darker = improvement/best per probe. The random control is omitted: it trained only to 500K, below this window.)

	GearNet	ProteinMPNN	ESM2
Baseline	0.6^{L6}	0.9^{L7}	2.7^{L8}
REPA L4-GearNet	$\mathbf{2.6^{L4}}$†	2.9^{L6}	8.2^{L8}
REPA L9-GearNet	$\mathbf{22.9^{L8}}$†	26.7^{L8}	40.1^{L8}
REPA L4-MPNN	4.8^{L7}	$\mathbf{16.6^{L7}}$†	19.8^{L5}
REPA L9-MPNN	5.0^{L7}	$\mathbf{14.2^{L6}}$†	14.2^{L7}

Table A.5: **Every REPA variant raises per-residue alignment to all three encoders, including the two it was never aligned to.** Across-layer peak per-residue CKNNA (bootstrap median, $\times 10^3$), with the peak trunk layer as a superscript; $k=10$, step 1.0M, $n \leq 256$ PDB-trained. The † marks each model's own alignment target. Every REPA row exceeds the baseline in every column, the Platonic-convergence signature discussed in §6.2.2. Absolute values are small (an order of magnitude below image-domain REPA), so we read the pattern, not the magnitude.

A.7 Representation–generation coupling on AFDB

Section 6.2.3 establishes the representation–generation coupling on PDB-trained models (Table 6.4). Table A.6 repeats the analysis for AFDB-trained models. The coupling holds here too, and in the same encoder-matched form. Local probes (inverse folding, dihedral) track designability, while the fold probe (CATH-A) tracks FPSD. The off-axis pairs are weak or absent, the signature of an encoder-routed effect rather than a generic one. We read AFDB designability with the proxy caveat of §6.1.1, so we lean on the FPSD column and the routing pattern rather than the absolute designability correlation.

Representation quality	FPSD		Designability	
	partial	raw	partial	raw
Inverse folding (top-1)	-0.17	-0.16	+0.50	+0.50
Dihedral MAE	+0.03	-0.03	+0.37	+0.09
CATH-A	+0.42	+0.43	+0.29	+0.34

Table A.6: **On AFDB too, better trunk representations track better generation.** The AFDB counterpart to Table 6.4: partial Pearson controlling for training step (raw alongside), oriented so positive means better representation accompanies better generation. Generation is FPSD-AFDB and designability; representation probes are read at n_{train}=1000 on the cross-database blinded set. (n=40 AFDB checkpoints pooled over the baseline, the REPA variants, and the random control.)

A.8 The ssJSD-2D secondary-structure metric

Proteina's secondary-structure metric compares the *mean* helix and strand content of a generated set against a reference. This collapses each set to a single point, so it cannot see variance, bimodality, or tails — a model that matches the average composition while collapsing its spread scores the same as one that reproduces it. ssJSD-2D removes this blind spot by comparing the full joint distribution.

For each backbone we annotate every residue as helix, strand, or coil with a Cα-only secondary-structure assignment (P-SEA, as implemented in Biotite) and record the fraction of residues in helix (f_H) and in strand (f_E). A set of N backbones is then a cloud of N points in the (f_H, f_E) plane. We bin that plane into a 5×5 grid over $[0, 1]^2$, form normalised two-dimensional histograms for the generated and reference sets, and report the Jensen–Shannon divergence between them (lower is better). Binning the joint distribution, rather than comparing marginal means, lets the metric reward a model for reproducing the *shape* of the secondary-structure distribution and penalise mode collapse onto, say, all-α backbones.

A.9 Secondary-structure composition of the designable subset

The encoder-routing reading in §6.2.5 — GearNet shifting the designable subset toward β-strand content, MPNN on AFDB away from it — rests on the secondary-structure composition of the designable backbones (Table A.7). The shifts are small in absolute terms but seed-consistent: on AFDB the L9-GearNet and L9-MPNN strand and helix ranges do not overlap across seeds, in opposite directions. The main-text E%des figures (Table 6.7) are the per-seed means of the strand column; we report the spread here rather than in the main table to avoid clutter.

Variant (700K, γ=0.45)	Strand f_E	Helix f_H	Seeds
PDB-trained			
Baseline	0.22 ± 0.02	0.48 ± 0.03	3
REPA L4-random	0.11 ± 0.01	0.61 ± 0.02	3
REPA L4-GearNet	0.22 ± 0.01	0.40 ± 0.02	3
REPA L9-GearNet	0.18 ± 0.02	0.48 ± 0.03	3
REPA L4-MPNN	0.23 ± 0.01	0.39 ± 0.03	3
REPA L9-MPNN	0.15 ± 0.01	0.54 ± 0.02	3
AFDB-trained			
Baseline	0.16 ± 0.01	0.47 ± 0.01	2
REPA L4-GearNet	0.14 ± 0.00	0.47 ± 0.01	2
REPA L9-GearNet	0.19 ± 0.01	0.38 ± 0.03	3
REPA L9-MPNN	0.11 ± 0.00	0.55 ± 0.01	2

Table A.7: **Secondary-structure composition of the designable subset (step 700K, γ=0.45).** Mean strand (f_E) and helix (f_H) fractions over the designable backbones, $\pm$ half the seed range. GearNet shifts the designable subset toward strand and away from helix; MPNN on AFDB does the reverse. The AFDB L9-GearNet vs. L9-MPNN ranges are non-overlapping on both axes. The strand column is the absolute form of the E%des column in Table 6.7, whose entries are the per-seed means reported here. (AFDB L4-MPNN omitted: no multi-seed γ=0.45 data.)

Encoder	**Dim**	**atom probe**	**context-Δ**	**3D-aware**	**Floor**	**Lift**
CheMeleon (2D graph)	2048	1.000	0.093	no	0.43	+0.04
MACE (3D potential)	192	1.000	0.260	weak	0.86	+0.005

Encoder	**Dim**	**Sparsity**	**RankMe**
CheMeleon (2D graph)	2048	93.8%	1166
MACE (3D potential)	192	0.0%	40.6

Table A.8: Full molecule encoder profile. *atom probe*: linear-probe accuracy for atom type (trivial in QM9, so omitted from the main table). *context*-Δ: same-element, different-environment cosine spread. *Floor*/*Lift* as in Table 4.2. RankMe is the Roy–Vetterli effective rank (GEOM value).

A.10 Full encoder profiles

Tables 4.2 and 4.3 in the main text show only the diagnostics that drive each verdict. Tables A.8 and A.9 give the complete set of measured values for every candidate encoder, including the columns omitted from the main text (atom/context probes, participation ratio, and the best-projector cosine the lift is computed from).

Encoder	**Dim**	**AA-probe**	**context-Δ**	**Floor**	**Best**	**Lift**	**RankMe**	**PR**
GearNet (CATH-trained)	512	0.137	0.222	0.425	0.434	+0.009	256.4	37.9
random-init	512	0.127	0.035	0.952	0.954	+0.003	99.8	1.5
ProteinMPNN (inv-fold)	128	0.327	0.022	0.835	0.850	+0.014	84.9	33.3
ESM2-650M (last layer)	1280	0.998	0.098	0.671	0.723	+0.053	1030.5	43.3
PW-GearNet (torsional)	3072	0.928	0.102	0.710	0.723	+0.013	207.7	5.5
MC-GearNet-Edge	3072	0.146	0.043	0.855	0.858	+0.002	12.0	1.0

Table A.9: Full protein encoder profile. *AA-probe*, *context-*Δ (within-minus-between protein cosine), *Floor* (mean-direction cosine), *Best* (best-projector cosine), *Lift* (= Best − Floor), *RankMe* and *PR* (participation ratio). MC-GearNet's mean L2 norm is 1.5×10^6, with 95% of its variance in a single dimension.